

# Relation Algebra

## §1. Notation

Traditionally Adjunction  $F \dashv G$  is defined as  $F(x) \leq y$  iff  $x \leq G(y)$ , and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define  $X : * \mapsto x$ ,  $Y : * \mapsto y$  and use diagram order, application can be replaced by composition:  $XF \leq Y$  iff  $X \leq YG$

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• write  $F$  as  $/$ ,  $G$  as  $\backslash$

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• render  $a \leq b$  as  $\boxed{\begin{smallmatrix} a \\ b \end{smallmatrix}}$

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• **rewrite**  $X \leq XF$  as  $1 \leq F$

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(1a)

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj

$$\begin{aligned} \boxed{\frac{X/}{X/}} &\xRightarrow{\text{adj}} \boxed{\frac{X}{X\backslash}} \xRightarrow{\text{rewrite}} \boxed{\frac{*}{\backslash}}_{\text{unit}} \xRightarrow{\text{put } \backslash \text{ on left}} \boxed{\frac{\backslash}{\backslash\backslash}} \\ &\xRightarrow{\text{put } / \text{ on right}} \boxed{\frac{/}{\backslash\backslash}} \\ \boxed{\frac{Y\backslash}{Y\backslash}} &\xRightarrow{\text{adj}} \boxed{\frac{Y\backslash}{Y}} \xRightarrow{\text{rewrite}} \boxed{\frac{\vee}{*}}_{\text{counit}} \xRightarrow{\text{put } / \text{ on left}} \boxed{\frac{\vee}{/}} \\ &\xRightarrow{\text{put } \backslash \text{ on right}} \boxed{\frac{\vee}{\backslash}} \end{aligned}$$

now we have  $\vee = /$  and  $\backslash\backslash = \backslash$ , and you just discovered string diagrams.

## §1.1. Adjunctions

(1.1a)

| $F \dashv G$                                              | $F$ 's type                                                           | monad $FG$                         | $1 \in FG$                                                                     | $GF \in 1$                                                                     | $F$ preserves $u$                                        | $G$ preserves $n$                                             | $FGF=F$                                                         | $GFG=G$                                                         |
|-----------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| $\triangleleft \dashv \triangleright$                     | $A \rightarrow A \otimes A$                                           | $\triangleleft \triangleright = 1$ | $1 \in \triangleleft \triangleright$                                           | $\triangleright \triangleleft \in 1$                                           | $(RuS) \triangleleft = R \triangleleft uS \triangleleft$ | $(RnS) \triangleright = R \triangleright nS \triangleright$   | $\triangleleft \triangleright \triangleleft = \triangleleft$    | $\triangleright \triangleleft \triangleright = \triangleright$  |
| $\multimap \dashv \multimap$                              | $A \rightarrow I$                                                     | $\multimap \multimap = T$          | $1 \in \multimap \multimap$                                                    | $\multimap \multimap \in 1$                                                    | $(RuS) \multimap = R \multimap uS \multimap$             | $(RnS) \multimap = R \multimap nS \multimap$                  | $\multimap \multimap \multimap = \multimap$                     | $\multimap \multimap \multimap = \multimap$                     |
| $\circ \dashv \circ$                                      | $(A \rightarrow B) \rightarrow (B \rightarrow A)$                     | $\circ \circ = 1$                  | $R = R \circ \circ$                                                            | $R = R \circ \circ$                                                            | $(RuS) \circ = R \circ uS \circ$                         | $(RnS) \circ = R \circ nS \circ$                              | $R \circ \circ \circ = R \circ$                                 | $R \circ \circ \circ = R \circ$                                 |
| $\multimap \triangleleft \dashv \triangleright \multimap$ | $(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$ | $1$                                | $(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$ | $(1 \otimes \multimap \triangleleft) (\triangleright \multimap \otimes 1) = 1$ | $=$                                                      | $=$                                                           | $=$                                                             | $=$                                                             |
| $\triangleright \multimap \dashv \multimap \triangleleft$ | $(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$ | $1$                                | $(1 \otimes \multimap \triangleleft) (\triangleright \multimap \otimes 1) = 1$ | $(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$ | $=$                                                      | $=$                                                           | $=$                                                             | $=$                                                             |
| $\Delta \dashv n$                                         | $(A \rightarrow B) \rightarrow (A \rightarrow B)^2$                   | $R \mapsto RnR = R$                | $R \in RnR$                                                                    | $RnS \in R$                                                                    | $\Delta(RuS) = \Delta Ru \Delta S$                       | $(RnT) n (SnU) = (RnS) n (TnU)$                               | $RnR = R$                                                       | $RnR = R$                                                       |
| $u \dashv \Delta$                                         | $(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$                   | $(R, S) \mapsto (RuS, RuS)$        | $R \in RuS$                                                                    | $RuR \in R$                                                                    | $(RuT) u (SuU) = (RuS) u (TuU)$                          | $\Delta(RnS) = \Delta Rn \Delta S$                            | $RuR = R$                                                       | $RuR = R$                                                       |
| $\perp \dashv !$                                          | $\{*\} \rightarrow (A \rightarrow B)$                                 | —                                  | —                                                                              | $\perp \in R$                                                                  | —                                                        | —                                                             | —                                                               | —                                                               |
| $\cdot S \dashv /S$                                       | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $S/S$                              | $R \in (RS)/S$                                                                 | $(T/S) S \in T$                                                                | $(RuT) S = RSuTS$                                        | $(RnT)/S = R/SnT/S$                                           | $((RS)/S) S = RS$                                               | $((T/S) S)/S = T/S$                                             |
| $S \cdot \dashv S \setminus$                              | $(B \rightarrow C) \rightarrow (A \rightarrow C)$                     | $S \setminus S$                    | $R \in S \setminus (SR)$                                                       | $S(S \setminus T) \in T$                                                       | $S(RuT) = SRuST$                                         | $S \setminus (RnT) = S \setminus RnS \setminus T$             | $S(S \setminus (SR)) = SR$                                      | $S \setminus (S(S \setminus T)) = S \setminus T$                |
| $Rn \dashv R \Rightarrow$                                 | $(A \rightarrow B) \rightarrow (A \rightarrow B)$                     | $X \mapsto R \Rightarrow (XnR)$    | $X \in R \Rightarrow (XnR)$                                                    | $Rn(R \Rightarrow Y) \in Y$                                                    | $Rn(XuY) = (RnX) u (RnY)$                                | $R \Rightarrow (XnY) = (R \Rightarrow X) n (R \Rightarrow Y)$ | $Rn(R \Rightarrow (XnR)) = XnR$                                 | $R \Rightarrow (Rn(R \Rightarrow Y)) = R \Rightarrow Y$         |
| $\mathcal{D} \dashv \cdot T$                              | $(A \rightarrow B) \rightarrow \text{Cor } A$                         | $R \mapsto (\mathcal{D}R)T$        | $R \in (\mathcal{D}R)T$                                                        | $\mathcal{D}(AT) \in A$                                                        | $\mathcal{D}(RuS) = \mathcal{D}Ru \mathcal{D}S$          | $(AnB)T = ATnBT$                                              | $\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$                   | $(\mathcal{D}(AT))T = AT$                                       |
| $\mathcal{R} \dashv T \cdot$                              | $(A \rightarrow B) \rightarrow \text{Cor } B$                         | $R \mapsto T(\mathcal{R}R)$        | $R \in T(\mathcal{R}R)$                                                        | $\mathcal{R}(TA) \in A$                                                        | $\mathcal{R}(RuS) = \mathcal{R}Ru \mathcal{R}S$          | $T(AnB) = TAnTB$                                              | $\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$                   | $T(\mathcal{R}(TA)) = TA$                                       |
| $\cdot f \dashv \cdot f^\circ$                            | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $ff^\circ$                         | $1 \in ff^\circ$                                                               | $f^\circ f \in 1$                                                              | $(RuS)f = RfuSf$                                         | $(RnS)f^\circ = Rf^\circ nSf^\circ$                           | $ff^\circ f = f$                                                | $f^\circ ff^\circ = f^\circ$                                    |
| $f^\circ \cdot \dashv f \cdot$                            | $(A \rightarrow C) \rightarrow (B \rightarrow C)$                     | $ff^\circ$                         | $1 \in ff^\circ$                                                               | $f^\circ f \in 1$                                                              | $f^\circ(XuY) = f^\circ Xu f^\circ Y$                    | $f(XnY) = fXn fY$                                             | $ff^\circ f = f$                                                | $f^\circ ff^\circ = f^\circ$                                    |
| $i \dashv E$                                              | $\text{Map} \hookrightarrow \text{Rel}$                               | $E$                                | $\frac{1}{\exists} : A \rightarrow E \quad A$                                  | $\exists : E \rightarrow B \quad B$                                            | —                                                        | —                                                             | $\frac{\exists}{\exists} = 1$                                   | $\frac{R}{\exists} \exists = R$                                 |
| $\cdot \exists \dashv \frac{\exists}{\exists}$            | $\text{Map}(A, E \rightarrow B) \rightarrow (A \rightarrow B)$        | $1$                                | $\frac{f \exists}{\exists} = f$                                                | $\frac{R}{\exists} \exists = R$                                                | —                                                        | —                                                             | $\frac{\frac{R}{\exists} \exists}{\exists} = f \exists$         | $\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$ |
| $\frac{\exists}{\exists} \dashv \cdot \exists$            | $(A \rightarrow B) \rightarrow \text{Map}(A, E \rightarrow B)$        | $1$                                | $\frac{R}{\exists} \exists = R$                                                | $\frac{f \exists}{\exists} = f$                                                | —                                                        | —                                                             | $\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$ | $\frac{f \exists}{\exists} \exists = f \exists$                 |

## §1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
\boxed{\begin{array}{c} X \\ f(R/T) \end{array}} \xLeftrightarrow[f^\circ \cdot \dashv f \cdot] \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} \xLeftrightarrow[\cdot T \dashv /T] \boxed{\begin{array}{c} (f^\circ X) T \\ R \end{array}} \\
\text{f a map} \\
\\
\xLeftrightarrow[\text{associativity}] \boxed{\begin{array}{c} f^\circ(X T) \\ R \end{array}} \xLeftrightarrow[f^\circ \cdot \dashv f \cdot] \boxed{\begin{array}{c} X T \\ f R \end{array}} \xLeftrightarrow[\cdot T \dashv /T] \boxed{\begin{array}{c} X \\ (f R)/T \end{array}} \\
\\
\xRightarrow[\text{indirect equality}] \boxed{\begin{array}{c} f(R/T) \\ (f R)/T \end{array}}
\end{array}$$

|                       | $\cdot T$                                   | $T \cdot$                                         | $\cdot g$                                         | $g^\circ \cdot$                                  | $Tn$                                                      | $^\circ$                                                                                                               | $\circ \triangleleft$                                                                                                  | $\Delta$                                                                | (1.2b) |
|-----------------------|---------------------------------------------|---------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------|
| $\cdot S$             | $R/(ST) = (R/T)/S$                          | $T \setminus (R/S) = (T \setminus R)/S$           | $R/(Sg) = (Rg^\circ)/S$                           | $g(R/S) = (gR)/S$                                | —                                                         | $(Y/S)^\circ = S^\circ \setminus (Y^\circ)$                                                                            | $(RS)^\wedge = R^\wedge (S \otimes 1)$                                                                                 | $(T_1 n T_2)/S = T_1/S n T_2/S$                                         |        |
| $S \cdot$             | $S \setminus (R/T) = (S \setminus R)/T$     | $(TS) \setminus R = S \setminus (T \setminus R)$  | $S \setminus (Rg^\circ) = (S \setminus R)g^\circ$ | $(g^\circ S) \setminus R = S \setminus (gR)$     | —                                                         | $(S \setminus Y)^\circ = Y^\circ/S^\circ$                                                                              | $((S \otimes 1)R)^\wedge = S \ R^\wedge$                                                                               | $S \setminus (T_1 n T_2) = S \setminus T_1 n S \setminus T_2$           |        |
| $\cdot f$             | $R/(fT) = (R/T)f^\circ$                     | $T \setminus (Rf^\circ) = (T \setminus R)f^\circ$ | $(fg)^\circ = g^\circ f^\circ$                    | —                                                | —                                                         | $(fY)^\circ = Y^\circ f^\circ$                                                                                         | —                                                                                                                      | $(T_1 n T_2)f^\circ = T_1 f^\circ n T_2 f^\circ$                        |        |
| $f^\circ \cdot$       | $f(R/T) = (fR)/T$                           | $(Tf^\circ) \setminus R = f(T \setminus R)$       | —                                                 | $(fg)^\circ = g^\circ f^\circ$                   | —                                                         | $(fY)^\circ = Y^\circ f^\circ$                                                                                         | —                                                                                                                      | $f(T_1 n T_2) = fT_1 n fT_2$                                            |        |
| $Rn$                  | —                                           | —                                                 | —                                                 | —                                                | $(RnT) \Rightarrow Y = R \Rightarrow (T \Rightarrow Y)$   | $(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$                                                              | —                                                                                                                      | $R \Rightarrow (T_1 n T_2) = (R \Rightarrow T_1) n (R \Rightarrow T_2)$ |        |
| $^\circ$              | $(Y/T)^\circ = T^\circ \setminus (Y^\circ)$ | $(T \setminus Y)^\circ = Y^\circ/T^\circ$         | $(gY)^\circ = Y^\circ g^\circ$                    | $(gY)^\circ = Y^\circ g^\circ$                   | $(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$ | $Y^\circ = Y$                                                                                                          | $(R^\wedge)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$<br>both bends: the $^\circ$ section's display | $(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$                             |        |
| $\circ \triangleleft$ | $(RT)^\wedge = R^\wedge (T \otimes 1)$      | $((T \otimes 1)R)^\wedge = T \ R^\wedge$          | —                                                 | —                                                | —                                                         | $(R^\wedge)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$<br>both bends: the $^\circ$ section's display | —                                                                                                                      | —                                                                       |        |
| $u$                   | $(X_1 u X_2)T = X_1 T u X_2 T$              | $T(X_1 u X_2) = TX_1 u TX_2$                      | $(X_1 u X_2)g = X_1 g u X_2 g$                    | $g^\circ(X_1 u X_2) = g^\circ X_1 u g^\circ X_2$ | $Tn(X_1 u X_2) = (TnX_1)u(TnX_2)$                         | $(X_1 u X_2)^\circ = X_1^\circ u X_2^\circ$                                                                            | —                                                                                                                      | —                                                                       |        |
| $\perp$               | $\perp T = \perp$                           | $T \perp = \perp$                                 | $\perp g = \perp$                                 | $g^\circ \perp = \perp$                          | $Tn\perp = \perp$                                         | $\perp^\circ = \perp$                                                                                                  | —                                                                                                                      | —                                                                       |        |

### §1.3. Adjoint triples

Beyond  $u \dashv \Delta \dashv n$ , which §1.1 already carries as two rows, the table holds one adjoint triple with content:  $\exists\_f \dashv f^* \dashv \forall\_f$  along a map  $f$ , once on each side of the composite.  $f : A \rightarrow B$  throughout this subsection.

$$\begin{aligned} \cdot f &\dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot &\dashv f \cdot \dashv f \setminus \end{aligned} \quad (1.3a)$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor:  $\cdot S \dashv /S$  at  $S := f^\circ$ , and  $S \cdot \dashv S \setminus$  at  $S := f$ .

The third link is not the first over again. For a map  $f$ ,  $/f$  collapses to  $\cdot f^\circ$ ; being a map is exactly what that collapse spends  $R/(fS) = (R/S)f^\circ$ . —  $f^\circ$  is not a map, so  $/f^\circ$  does not collapse in turn, and the three operators stay distinct. On  $\text{Rel}(\mathbb{I}, A) = E\ A$  they are the image triple:

| operator        | acts by                                   | name                       | (1.3b) |
|-----------------|-------------------------------------------|----------------------------|--------|
| $\cdot f$       | $S \mapsto f[S]$                          | direct image, $\exists\_f$ |        |
| $\cdot f^\circ$ | $T \mapsto f^{-1}[T]$                     | inverse image, $f^*$       |        |
| $/f^\circ$      | $S \mapsto \{b : f^{-1}(b) \subseteq S\}$ | $\forall\_f$               |        |

§1.1's  $\mathcal{D} \dashv \cdot T$  is this same chain along the projection  $A \otimes B \rightarrow A$ , read through  $\text{Rel}(A, B) = E(A \otimes B)$ :  $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$ ,  $AT$  is the relation that ignores  $b$  altogether, and the third link is  $R \mapsto 1 \cap R/T = \{(a, a) : \forall b. aRb\}$ .

$$\mathcal{D} \dashv \cdot T \dashv 1n\cdot/T \quad (1.3c)$$

Three is where it stops, and one map breaks both ends. Take  $A = \{a_1, a_2\}$  and  $B = \{b\}$ :  $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$  while  $f[\{a_1\} \cap \{a_2\}] = \emptyset$ , so  $\cdot f$  does not preserve meets and has no left adjoint, and the same two subsets give  $\forall\_f(\{a_1\} \cup \{a_2\}) = \{b\}$  against  $\forall\_f\{a_1\} \cup \forall\_f\{a_2\} = \emptyset$ , so  $/f^\circ$  does not preserve joins and has no right adjoint. A monic  $f$  restores the binary case and no more —  $\tau f$  still reaches only  $\text{im } f$ , and  $\forall\_f \perp = B \setminus \text{im } f$  is still not  $\perp$ . The chain extends only when  $f^\circ$  is a map as well, and then  $/f^\circ = \cdot f^{\circ\circ} = \cdot f$  and it repeats forever. For an  $S$  with neither  $S$  nor  $S^\circ$  a map there is no triple at all:  $\cdot S \dashv /S$  is two links and stops.

The rows that chain forever say nothing by it:  $^\circ$  is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and  $^\circ \dashv \circ \dashv \circ$  has no end, and §1.1's other row of that kind,  $\circ \triangleleft \dashv \triangleright \circ$ , goes the same way.

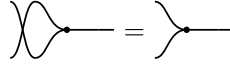
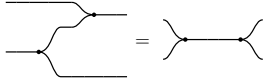
## §2. Relations

$\mathbf{Rel}$  is a poset-enriched category with  $(\otimes, \circ)$  where  $\otimes$  is commutative cartesian product, and converse  $\circ \dashv \circ$ , and

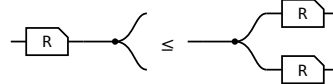
$\mathbf{C}: (\triangleright : A \otimes A \rightarrow A, \circ : \mathbb{I} \rightarrow A)$  a commutative monoid with  $\mathbf{C}^\circ \dashv \mathbf{C}$ . We write  $\mathbf{C}^\circ$  as  $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow \mathbb{I})$



$\triangleright$  associative

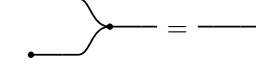


$\triangleright$  commutative

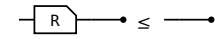


Frobenius, one half — the other is its  $\circ$

$R \triangleleft \leq \triangleleft (R \otimes R)$



$\circ$  is its unit

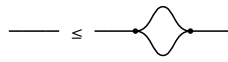


$R \circ \leq \circ$

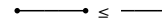
### §2.1. $\forall$ object $\mathbf{A}$ , $(\mathbf{A}, \triangleleft, \circ) \dashv (\mathbf{A}, \triangleright, \circ)$



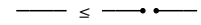
$\triangleright \triangleleft \leq 1$



$1 \leq \triangleleft \triangleright$



$\circ \circ \leq 1$



$1 \leq \circ \circ$

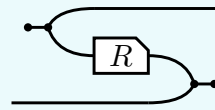
(2.1a)

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In  $\mathbf{Rel}$  it reads  $\{(a, a)\} \subseteq A \times A$  — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

### §3. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$^\circ$  is primitive, part of the data the first section lists. It turns both of  $R$ 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are  $R^\circ$ , not its definition:



$$R^\circ = (\circlearrowleft \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circlearrowright)$$

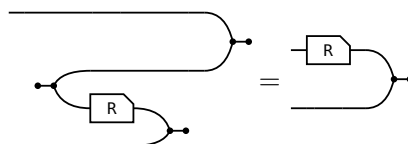
where  $\circlearrowleft \triangleleft : \mathbb{I} \rightarrow a \otimes a$  opens a pair of wires out of nothing and  $\triangleright \circlearrowright : b \otimes b \rightarrow \mathbb{I}$  closes one, so the input of  $R^\circ$  is where the output of  $R$  was. Taking that formula as the definition would now be circular:  $\triangleleft$  and  $\circlearrowright$  are  $\triangleright^\circ$  and  $\circlearrowleft^\circ$ . The laws of  $^\circ$  are that it is a contravariant 2-functor  $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ :

$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

#### §3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:

(3.1a)

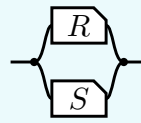


$R^\circ$  is DEFINED as the bending of  $(R \otimes 1) \triangleright \circlearrowright$ , so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the  $\circlearrowleft \triangleleft$  bends the  $a$  strand down and the  $\triangleright \circlearrowright$  brings it back up, while the  $b$  strand rides through untouched. Nothing else is spent below.

## §4. $\cap$ is a commutative idempotent monoid on every hom-set

(4a)

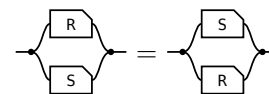
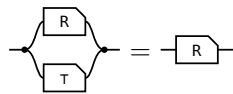
The **meet** of  $R, S : a \rightarrow b$ , the paper's **convolution**, is  $R \cap S := \triangleleft (R \otimes S) \triangleright$  — copy the input, run  $R$  and  $S$  on the two copies, merge the results — so what comes out is what both of them do.



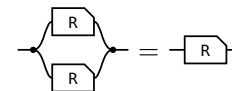
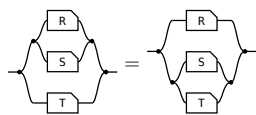
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow  $\tau = \multimap \multimap$  (the paper's Lemma 4.11).

(4b)



**unit:** one half of  $\tau$  per end — the merge's unit law absorbs the  $\multimap$ , the copy's counit law the  $\multimap$  **commutative:**  $\sigma$  crosses  $R \otimes S$  by naturality and is absorbed by cocommutativity and commutativity



**associative:** coassociativity and associativity;  $\otimes$  re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So  $\leq$  is the order this monoid induces.  $R \cap S \leq R$  comes from the unit, and idempotency turns anything under both  $S$  and  $T$  into something under  $S \cap T$ , since  $R = R \cap R \leq S \cap T$ .

And one law relating  $\cap$  to composition, which is **not** an equation:

(4c)

| semi-distributivity, and what supplies it                                                                                                                        | picture |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $R (S \cap T) \sqsubseteq R S \cap R T$ — the lax copy law.<br>Equality exactly when $R$ is single valued: the Maps section's<br>$F (R \cap S) = F R \cap F S$ . |         |

## §5. Domain and range

The **domain**  $\text{Dom } R \triangleq 1 \cap R R^\circ$  and the **range**  $\text{Ran } R \triangleq \text{Dom } R^\circ$ .



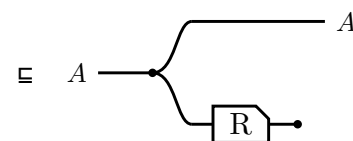
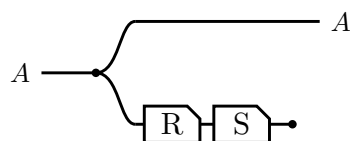
$\triangleright$  lands the return leg back on the value  $\triangleleft$  handed out,  
so  $R^\circ \triangleright$  cuts to  $\circ$  — Frobenius

Running  $R$  and throwing the result away leaves only the fact that  $R$  could fire, and  $\text{Ran } R$  is the same picture with the box mirrored. In  $\text{Rel}$  both steps are  $\{(a, a) : \exists b. a R b\}$ .

|                                                                                            |
|--------------------------------------------------------------------------------------------|
| $\text{Dom } R \triangleq 1 \cap R R^\circ$                                                |
| $\text{Dom } R \subseteq A \iff R \subseteq A R, \text{ for } A \text{ coreflexive}$       |
| $\text{Dom } (R S) \subseteq \text{Dom } R$                                                |
| $\text{Dom } (R \cap S) = 1 \cap S R^\circ$                                                |
| $R \text{ entire} \iff \text{Dom } R = 1 \iff 1 \subseteq R R^\circ$                       |
| $R \text{ simple} \iff R^\circ R \subseteq 1$                                              |
| $R \text{ a map} \iff R \text{ entire and simple}$                                         |
| $R, S \text{ entire} \implies R S \text{ entire} \text{ — likewise simple, likewise maps}$ |
| $R S \text{ entire} \implies R \text{ entire}$                                             |

### §5.1. Sliding the discard

$\text{Dom } (R S) \subseteq \text{Dom } R$ , and a single glyph for  $\text{Dom}$  would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

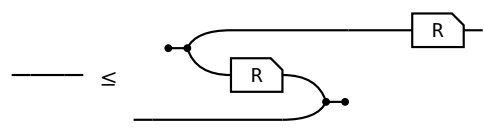
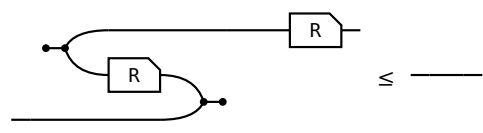
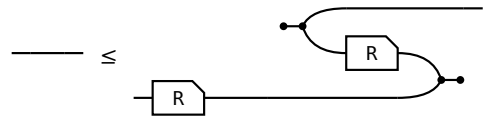
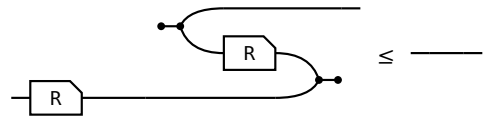


$S \circ \circ \subseteq \circ$ , the lax axiom for  $\circ$  in the first section  
— the discard slides back past  $S$

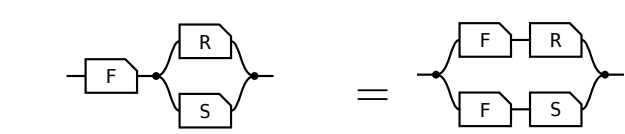
Equality is  $S \text{ entire}$ , which is the same picture read as  $\text{Dom } R = 1 \iff R \text{ entire}$ .

## §6. Maps

Every arrow is lax for  $\triangleleft$  and for  $\multimap$  by axiom, and for  $\triangleright$  and  $\multimap$  by taking  $\circ$  of those two. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

|                                                                                                                                                                                             |                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>surjective</b><br/> <math>1 \sqsubseteq R^\circ R</math>, that is, <math>R^\circ</math> entire.</p> |  <p><b>single valued</b><br/> <math>R^\circ R \sqsubseteq 1</math></p> |
|  <p><b>entire</b><br/> <math>1 \sqsubseteq R R^\circ</math>. With single valued, a map.</p>                |  <p><b>injective</b><br/> <math>R R^\circ \sqsubseteq 1</math></p>     |

(6a)

| law                                                                                                                                                                                                                                                                              | picture                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><math>F (R \cap S) = F R \cap F S</math></p> <p><math>F</math> single valued</p> <p><math>F</math> = people <math>x</math> may ask, <math>R</math> = admires <math>a</math>, <math>S</math> = admires <math>b</math>. Single valued means one person, so the sides agree.</p> |  <p>one person who admires both      <math>a</math> at <math>A</math>, <math>b</math> at <math>B</math></p> |

(6b)



## §7. / is all of

(7a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/ compares images:  $S(y) \subseteq R(x)$ .  $\setminus$  compares preimages:  $S^\circ(x) \subseteq R^\circ(y)$ .

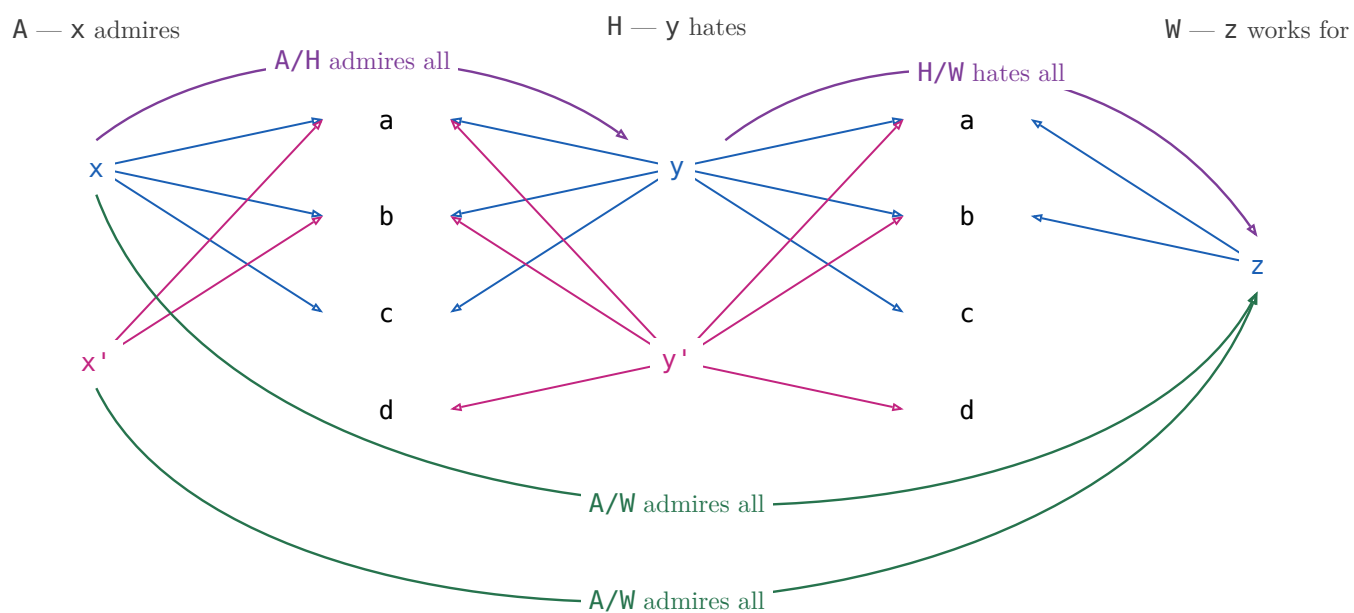
example: A admires, H hates, W works for

$x (A/H) y$  —  $x$  admires everyone  $y$  hates.

$x (H \setminus A) y$  — everyone who hates  $x$  admires  $y$ .

§7.1.  $(R/S)(S/W) \sqsubseteq R/W$ 

(7.1a)



$$x (A/H) y \text{ — } x \text{ admires everyone } y \text{ hates}$$

$$y (H/W) z \text{ — } y \text{ hates everyone } z \text{ works for}$$

$$x (A/W) z \text{ — } x \text{ admires everyone } z \text{ works for}$$

(7.1b)

$(A/H)(H/W)$  is a path:  $x \rightarrow y \rightarrow z$ , and that is all of it.  $A/W$  also holds of  $x'$ , who admires everyone  $z$  works for — but  $x'$  does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of  $(R/S)(S/W) \sqsubseteq R/W$ .

|                                                                                                                                                                                                                               |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $X \sqsubseteq R/S \iff X S \sqsubseteq R$<br>$X$ is any $X$ -to- $Y$ pairing; one that only pairs $X$ with a $y$ such that $x$ admires everyone $y$ hates lies inside $R/S$ , and $R/S$ is the largest such.                 |  |
| $X \sqsubseteq S \setminus R \iff S X \sqsubseteq R$<br>The mirror — divide on the left when $X$ comes first.                                                                                                                 |  |
| $(R/S) S \sqsubseteq R$<br>There is a $y$ such that $x$ admires everyone $y$ hates, and $p$ is one of the people $y$ hates — then $x$ admires $p$ too. Strict at $S = \emptyset$ : $R/S$ is everyone, $(R/S) S = \emptyset$ . |  |
| $S (S \setminus R) \sqsubseteq R$<br>The mirror.                                                                                                                                                                              |  |
| <b>associate:</b> $R/(S_1 S_2) = (R/S_2)/S_1$<br>A friend's enemy is two hops: divide by the far end first.                                                                                                                   |  |
| $(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$<br>The mirror.                                                                                                                                                      |  |
| <b>maps:</b> $f (R/S) = (f R)/S$<br>Rename $x$ before or after dividing — the licence to write $f R / S$ .                                                                                                                    |  |
| $R/(f S) = (R/S) f^\circ$<br>Rename $y$ : a map leaves a denominator as $f^\circ$ outside the box.                                                                                                                            |  |
| $(R/S) (S/W) \sqsubseteq R/W$<br>Someone who admires all of a hate-set that already covers everyone $Z$ works for admires those people too.                                                                                   |  |
| $1 \sqsubseteq R/R$<br>$R/R$ runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only $a$ and $b$ admire each other's idols too, and still stay two people.                        |  |
| $(R/R) (R/R) = R/R$<br>$R/R$ is the preorder <b>admires at least as much as</b> , and a preorder is idempotent. Freyd writes $\sqsubseteq$ ; with $1 \sqsubseteq R/R$ above it is an equality.                                |  |
| $(R/R) R = R$<br>Reaching $p$ through someone whose idols $x$ fully admires is reaching $p$ directly, since $x$ admires their own idols.                                                                                      |  |
| $R/1 = R$<br>Dividing by $1$ : $p$ 's set is just $\{p\}$ , so admiring all of it is admiring $p$ .                                                                                                                           |  |
| $R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$<br>Admiring a combined hate-set is admiring each set in full.                                                                                                                           |  |
| $S \setminus (R/W) = (S \setminus R)/W$<br>Which is why $S \setminus R/W$ needs no bracket.                                                                                                                                   |  |

Fifteen laws, fifteen pictures, and not one shows a generator:  $\cap$ ,  $\cup$ ,  $\circ$  and composition are what the Frobenius generators build, and  $/$  is none of those — it is posited, with nothing to unfold.

# §8. $\frac{R}{S}$

$\frac{R}{S} \triangleq (R/S) \cap (S/R)^\circ$ . In Rel  $x$  and  $y$  has the same image:  $\forall p. (x R p \iff y S p)$

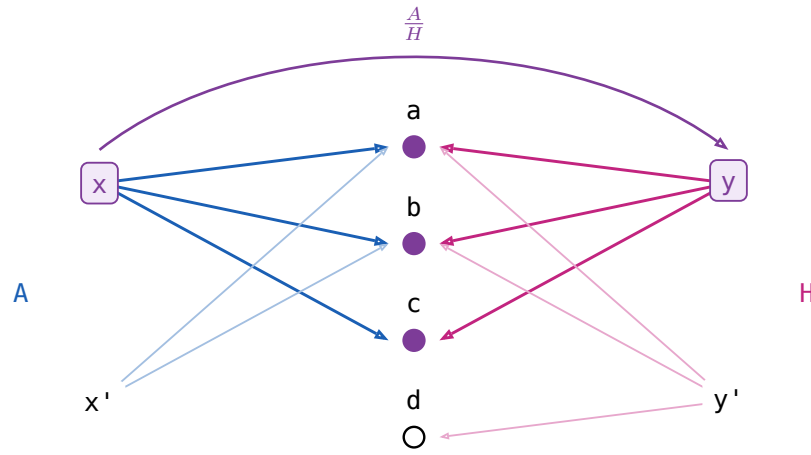
(8a)

$x (A/H) y$  —  $x$  admires everyone  $y$  hates  
 $x ((H/A)^\circ) y$  —  $x$  admires only people  $y$  hates  
 $x ((A/H) \cap (H/A)^\circ) y$  —  $x$  admires only and all whom  $y$  hates

(8b)

$x$  admires exactly whom  $y$  hates:  $/$  is **all of**,  $\frac{R}{S}$  is **only all of**.

(8c)



| law                                                                              | the reading                                                                                                                                        |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| $X \subseteq \frac{R}{S} \iff XS \subseteq R \text{ and } X^\circ R \subseteq S$ | $X$ may pair $x$ with $y$ only when $x$ admires exactly whom $y$ hates. Both halves must typecheck, so the operation is <b>partial</b> .           |
| $(\frac{R}{S})^\circ = \frac{S}{R}$                                              | Matching is symmetric.                                                                                                                             |
| $\frac{R}{S} \frac{S}{W} \subseteq \frac{R}{W}$                                  | And transitive.                                                                                                                                    |
| $\frac{R}{S} S \subseteq R$                                                      | $(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$<br>$x$ <b>only</b> admires whom $y$ hates<br>$\frac{R}{S} S = \text{Dom}(\frac{R}{S}) R$ |
| $\frac{R}{R} R = R$                                                              | $(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$<br>$x$ and $y$ admire the same people<br>$y = x$ always qualifies ( $1 \subseteq R\%R$ below)   |
| $1 \subseteq \frac{R}{R}$                                                        | $x(\frac{R}{R})y$ if $x$ and $y$ admires the same peoples.                                                                                         |
| $(\frac{R}{R})^2 = \frac{R}{R}$                                                  | So the relation <b>admires the same people</b> is an equivalence relation.                                                                         |
| $X \subseteq \frac{R}{R} \iff XR \subseteq R$ , for symmetric $X$                | The largest symmetric arrow that leaves $R$ alone.                                                                                                 |
| $\frac{R}{1}$ is the <b>simple part</b> of $R$                                   | The people who admire exactly one person and nobody else. It equals $R$ only when $R$ is simple, unlike $R/1 = R$ .                                |
| $\text{Dom } \frac{R}{S} = 1 \cap (R/S)(S/R)$                                    | Its domain is the <b>domain of simplicity</b> of $R$ .                                                                                             |

(8d)

## §9. Freyd's Power allegories

(9a)

A **power allegory** is a division allegory with one unary operation on arrows,  $\exists$  **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, \quad \exists_R = \exists_R \square \\ \mathbb{1} &\subseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ \frac{\exists_R}{\exists_R} &= \mathbb{1} & \exists_R \text{ is } \mathbf{straight} \end{aligned}$$

$R\square$  is  $R$ 's target, an identity arrow. For  $R : A \rightarrow B$  write  $\exists : E B \rightarrow B$ , dropping the subscript.

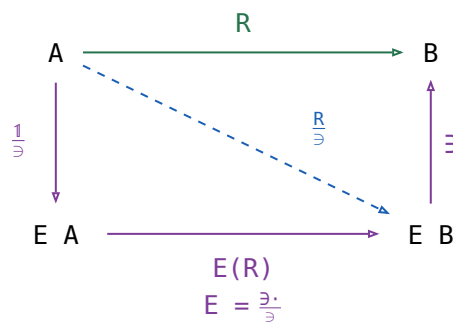
$$\epsilon \triangleq \exists^\circ : A \rightarrow E A$$

(9b)

| law                                                                                             | the reading                                                                                                                         |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1 $\exists_R \square = R\square, \exists_R = \exists_R \square$                                 | One $\exists$ per object, not per arrow.                                                                                            |
| 2 $\exists$ is <b>thick</b>                                                                     | <b>Comprehension:</b> every $x$ has a set of exactly the people $x$ admires.                                                        |
| 3 $\frac{R}{\exists} : A \rightarrow E B$ , for $R : A \rightarrow B$                           | convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$                                                          |
| 4 $\frac{R}{\exists}$ is a <b>map</b>                                                           |                                                                                                                                     |
| 5 $\frac{R}{\exists} \exists = R$                                                               |                                                                                                                                     |
| 6 $F \subseteq \frac{F \exists}{\exists}$ , $F$ simple                                          | A partial choice of sets is inside the total one.                                                                                   |
| 7 $\frac{1}{\exists}$ , the <b>singleton map</b> , monic                                        | The one-person set.                                                                                                                 |
| 8 $\frac{\exists}{\exists} = \mathbb{1}$                                                        | Make the set of a set, then read it back one level down.                                                                            |
| 9 <b>fusion:</b> $\frac{f R}{\exists} = f \frac{R}{\exists}$ , $f$ a map                        | Naturality of the unit, $f \frac{1}{\exists} = \frac{1}{\exists} (E(f))$ .                                                          |
| 10 $\frac{f}{\exists} = f \frac{1}{\exists}$ , $f$ a map                                        | Rename first or take singletons first — the fusion row above at $R = \mathbb{1}$ .                                                  |
| 11 $\frac{R}{S} = \frac{R}{\exists} \left( \frac{S}{\exists} \right)^\circ$                     | $x$ and $y$ match when $R$ sends $x$ and $S$ sends $y$ to the same set.                                                             |
| 12 $E(R) \triangleq \frac{\exists R}{\exists}$ , $E \triangleq \frac{\exists \cdot}{\exists}$   | $E(R) : E A \rightarrow E B$ , image of a set of $A$                                                                                |
| 13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$                                                 | $\frac{1}{\exists} : x \mapsto \{x\}$                                                                                               |
| 14 $\frac{S}{\exists} E(R) = \frac{S}{\exists} \frac{\exists R}{\exists} = \frac{S R}{\exists}$ | absorption                                                                                                                          |
| 15 <b>subset</b> $\triangleq \exists/\exists : E A \rightarrow E A$                             | $xs \text{ subset } ys \iff \forall a. ys \exists a \rightarrow xs \exists a$ , that is $ys \subseteq xs$ , not $xs \subseteq ys$ . |

### §9.1. $i \dashv E$ Power Allegory defined as adjunction

(9.1a)



$$\frac{R}{\exists} \exists = R \quad E A \text{ is the powerset of } A. \text{ B\&dM write } P A \text{ — standard mathematics, but here } P \text{ is already the relator } P(R).$$

## §10. Relator

(10a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from  $R$  to  $S$  IS  $R \subseteq S$ . A **relator**  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a 2-functor between allegories:

$$F(1) = 1 \quad F(R \circ S) = F(R) \circ F(S) \quad R \subseteq S \implies F(R) \subseteq F(S)$$

Preserving  $\circ$  is **not** asked for —  $\circ$  is an identity-on-objects involution  $\mathcal{C}^{\circ\circ} \rightarrow \mathcal{C}$ , no part of the 2-category.

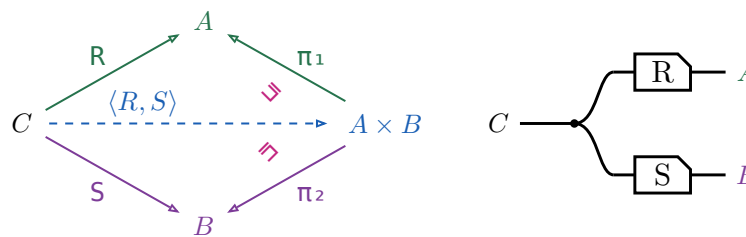
| the statement                                                                       | B&dM                      | (10b) |
|-------------------------------------------------------------------------------------|---------------------------|-------|
| For $f$ a map, $F(f)$ is a map and $F(f^\circ) = F(f)^\circ$ .                      | Lemma 5.1                 |       |
| Over a <b>tabular</b> allegory a functor is a relator $\iff$ it preserves $\circ$ . | Theorem 5.1               |       |
| $F(R^\circ) = F(R)^\circ$ for every $R$ , so $F(R)^\circ$ needs no bracket.         | after Theorem 5.1, p. 113 |       |
| Two relators agreeing on maps are equal.                                            | Corollary 5.1             |       |
| $F(X \cap Y) = F(X) \cap F(Y)$ for $X, Y$ coreflexive.                              | Ex 5.2                    |       |
| $F(R \cap S) \subseteq F(R) \cap F(S)$ , and strictly.                              | Ex 5.2, the restriction   |       |
| $F(\text{Dom } R) = \text{Dom}(F(R))$ for $F$ preserving $\circ$ .                  | —                         |       |

The **power relator**  $P \multimap xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$  — is where the fourth is strict: for  $R = \{(a_1, b_1), (a_2, b_2)\}$  and  $S = \{(a_1, b_2), (a_2, b_1)\}$  the pair  $(\{a_1, a_2\}, \{b_1, b_2\})$  is in  $P(R) \cap P(S)$ , while  $R \cap S = \emptyset$ .

## §11. Fork $\langle R, S \rangle$

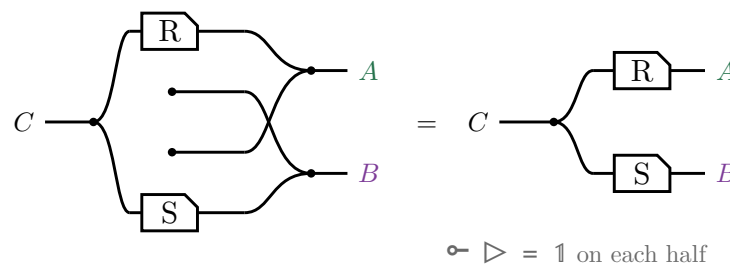
The **fork** of  $R : C \rightarrow A$  and  $S : C \rightarrow B$  is  $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$ , where  $(\pi_1, \pi_2)$  is the tabulation of  $\tau$ .

$$\langle R, S \rangle \pi_1 = (\text{Dom } S) R \quad \langle R, S \rangle \pi_2 = (\text{Dom } R) S$$



A domain is coreflexive, so  $\langle R, S \rangle \pi_1 \subseteq R$ , with equality exactly when  $S$  is entire; for maps both triangles commute and  $\langle f, g \rangle$  is unique. In **Rel**,  $c \ \langle R, S \rangle \ (a, b)$  iff  $c \ R \ a$  and  $c \ S \ b$  — copy  $c$ , then  $R$  on one strand and  $S$  on the other, which is  $\triangleleft (R \otimes S)$  on the right.

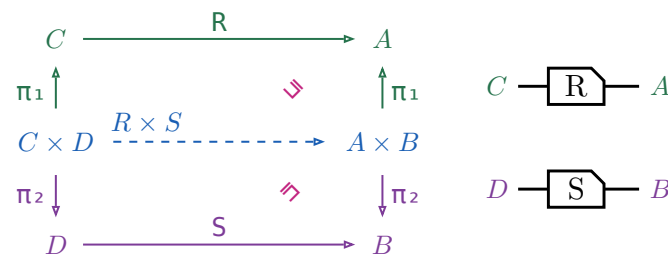
No  $\circ$  survives the translation.  $\pi_1 = 1 \otimes \multimap$  discards the second component, so  $\pi_1^\circ = 1 \otimes \multimap$  **creates** one out of nothing, and  $\cap$  is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



### §11.1. Relational product $R \times S$

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$ , a relator in each argument but no longer a categorical product.

(11.1b)

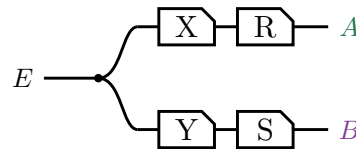


Right-then-up is  $(R \times S) \pi_1$ , up-then-right is  $\pi_1 R$ , and  $(R \times S) \pi_1 \sqsubseteq \pi_1 R$ , equality when  $S$  is entire. In  $\mathbf{Rel}$ ,  $(c, d) (R \times S) (a, b)$  iff  $c R a$  and  $d S b$  — two strands side by side, no copy dot:  $R \times S = R \otimes S$ .

## §11.2. Absorption

For  $X : E \rightarrow C$  and  $Y : E \rightarrow D$ ,  $\langle X, Y \rangle (R \times S) = \langle X R, Y S \rangle$ . Both sides are this picture:

(11.2a)



## §12. Coproduct $[R, S]$

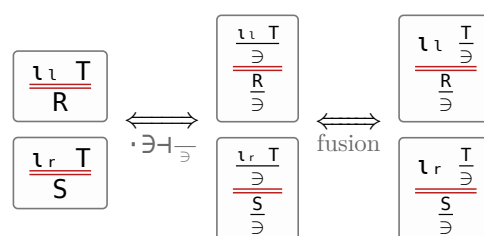
(12a)

| the statement                                                                                                                                                                                                  | picture |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $[R, S] \triangleq \iota_l^\circ R \cup \iota_r^\circ S$<br>The tape is the union — a particle entering at $A + B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint. |         |
| $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$                                                                                                                                                      |         |
| $R + S \triangleq [R \iota_l, S \iota_r]$                                                                                                                                                                      |         |
| $\iota_l [R, S] = R$ , $\iota_r [R, S] = S$ , and $[R, S]$ is the only such arrow                                                                                                                              |         |
| $\iota_l \iota_l^\circ = \mathbb{1} = \iota_r \iota_r^\circ$                                                                                                                                                   |         |
| $\iota_l \iota_r^\circ = \perp = \iota_r \iota_l^\circ$                                                                                                                                                        |         |
| $\iota_l^\circ \iota_l \cup \iota_r^\circ \iota_r = \mathbb{1}$                                                                                                                                                |         |
| $[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$                                                                                                                                                               |         |

### §12.1. $[R, S] \triangleq [\frac{R}{\exists}, \frac{S}{\exists}] \exists$

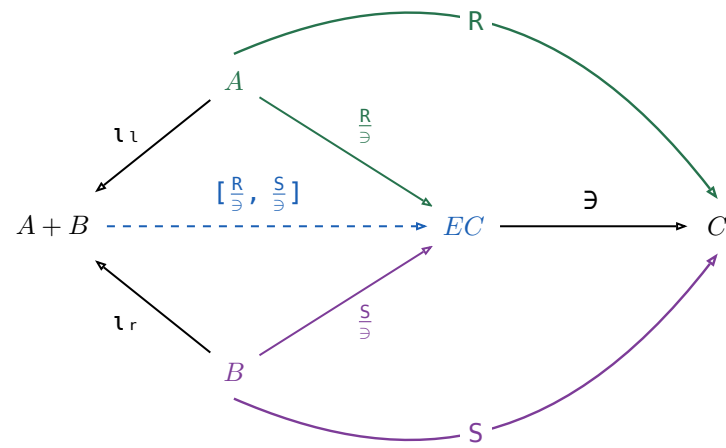
The universal-property row is not free:  $\iota_l$ ,  $\iota_r$  were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because  $\frac{\_}{\exists}$  sends an arrow  $A \rightarrow C$  to a map  $A \rightarrow E \rightarrow C$  reversibly, so the map coproduct can be applied underneath it. For any  $T : A + B \rightarrow C$ ,

(12.1a)



$$\begin{array}{ccc}
 \longleftrightarrow & \boxed{\begin{array}{c} \frac{T}{\exists} \\ \hline [\frac{R}{\exists}, \frac{S}{\exists}] \end{array}} & \xleftrightarrow{\cdot \exists \vdash \frac{\cdot}{\exists}} \boxed{\begin{array}{c} \frac{T}{\exists} \\ \hline [\frac{R}{\exists}, \frac{S}{\exists}] \exists \end{array}}
 \end{array}$$

(12.1b)



Nothing here holds only up to  $\Xi$ : every triangle commutes on the nose, which is the difference from the fork above. The border spells  $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$ , and pushing  $\exists$  into the union that  $[\cdot, \cdot]$  on maps already is turns that back into the definition,  $(1_l \circ \frac{R}{\exists} \cup 1_r \circ \frac{S}{\exists}) \exists = 1_l \circ R \cup 1_r \circ S$ .



# §13. The power relator $P(R)$

For  $R : A \rightarrow B$ ,

$$P(R) \triangleq ((\exists R)/\exists) \cap ((\exists R^\circ)/\exists)^\circ : E A \rightarrow E B$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R)/\exists) \cap (\exists/(\exists R))^\circ$$

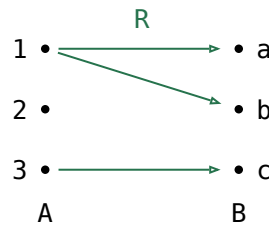
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of  $xs$  is related by  $R$  to some element of  $ys$ , and conversely.

(13a)

|                                     | in Rel                                                                                                 | in words                                                                          | in short                                                            |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------|
| $(\exists R)/\exists$               | $\forall y \in ys. \exists x \in xs. x \ R \ y$                                                        | $\forall y. \text{ some } xs \ R \ y$                                             | $ys$ has only what $xs$ reaches                                     |
| $(\exists/(\exists R))^\circ$       | $\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$                                        | every $y$ with some $xs \ R \ y$ is in $ys$                                       | $ys$ has all that $xs$ reaches                                      |
| $E(R)$                              | $ys = \{y \mid \exists x \in xs. x \ R \ y\}$                                                          | $ys = \text{every } y \text{ with some } xs \ R \ y$                              | $ys$ has all and only what $xs$ reaches                             |
| $((\exists R^\circ)/\exists)^\circ$ | $\forall x \in xs. \exists y \in ys. x \ R \ y$                                                        | $\forall x. x \ R \ \text{some } ys$                                              | every $x$ reaches into $ys$                                         |
| $P(R)$                              | $\forall y \in ys. \exists x \in xs. x \ R \ y$ and<br>$\forall x \in xs. \exists y \in ys. x \ R \ y$ | $\forall y. \text{ some } xs \ R \ y$ and<br>$\forall x. x \ R \ \text{some } ys$ | $ys$ has only what $xs$ reaches, and<br>every $x$ reaches into $ys$ |

(13b)



(13c)

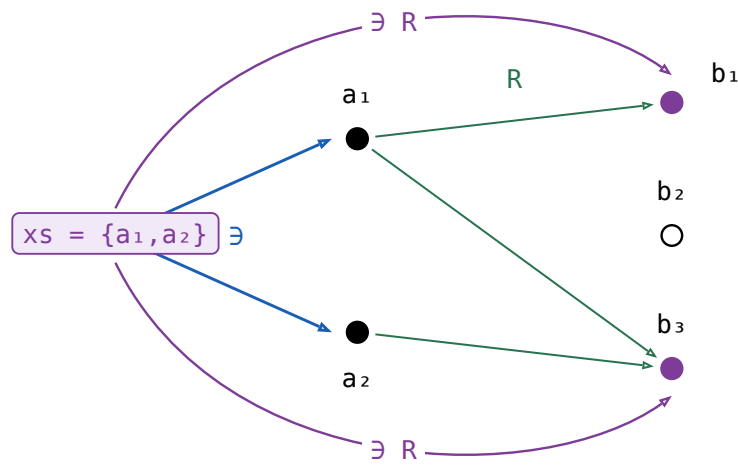
| $xs$          | $(\exists R)/\exists$  | $(\exists/(\exists R))^\circ$ | $E(R)$      | $((\exists R^\circ)/\exists)^\circ$ | $P(R)$        |
|---------------|------------------------|-------------------------------|-------------|-------------------------------------|---------------|
| $\emptyset$   | $\emptyset$            | all 8 subsets of $abc$        | $\emptyset$ | all 8 subsets of $abc$              | $\emptyset$   |
| $\{1\}$       | $\emptyset, a, b, ab$  | $ab, abc$                     | $ab$        | $a, b, ab, ac, bc, abc$             | $a, b, ab$    |
| $\{2\}$       | $\emptyset$            | all 8 subsets of $abc$        | $\emptyset$ | none                                | none          |
| $\{3\}$       | $\emptyset, c$         | $c, ac, bc, abc$              | $c$         | $c, ac, bc, abc$                    | $c$           |
| $\{1, 2\}$    | $\emptyset, a, b, ab$  | $ab, abc$                     | $ab$        | none                                | none          |
| $\{1, 3\}$    | all 8 subsets of $abc$ | $abc$                         | $abc$       | $ac, bc, abc$                       | $ac, bc, abc$ |
| $\{2, 3\}$    | $\emptyset, c$         | $c, ac, bc, abc$              | $c$         | none                                | none          |
| $\{1, 2, 3\}$ | all 8 subsets of $abc$ | $abc$                         | $abc$       | none                                | none          |

$R(2) = \emptyset$ , so  $P(R)$  sends no  $ys$  at all to any  $xs$  containing 2 — not entire.

$R(1)$  has two elements, so  $\{1\}$  gets three: 1 need only meet  $ys$ , not be swallowed by it — not simple.

Every  $E(R)$  row has exactly one  $ys$ , which is what makes it a map.

(13d)



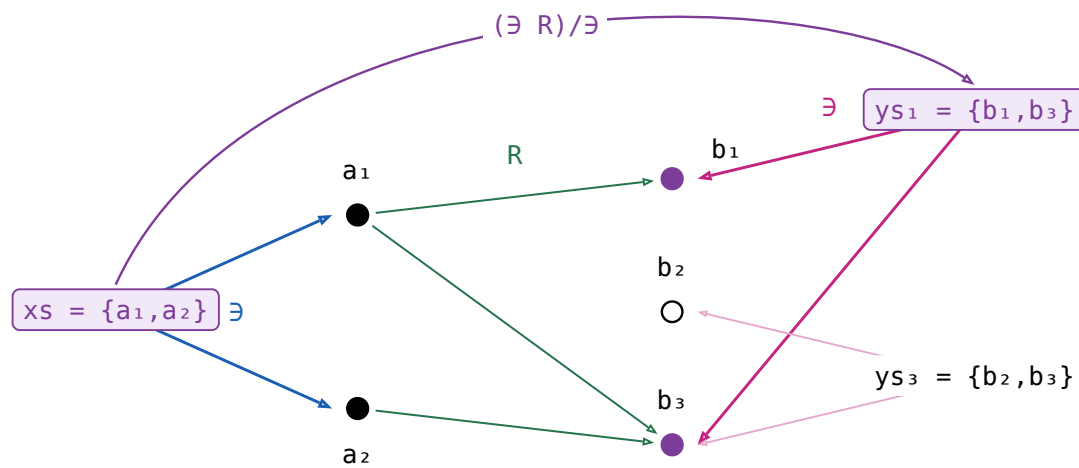
$$\exists R : E A \rightarrow B$$

$$xs \ (\exists R) \ b \iff \exists a \in xs. a \ R \ b$$

one arc per element of  $B$  that  $xs$  reaches, and  $b_2$  gets none

Dividing by  $\exists$  turns that into a relation between **sets**.

(13e)

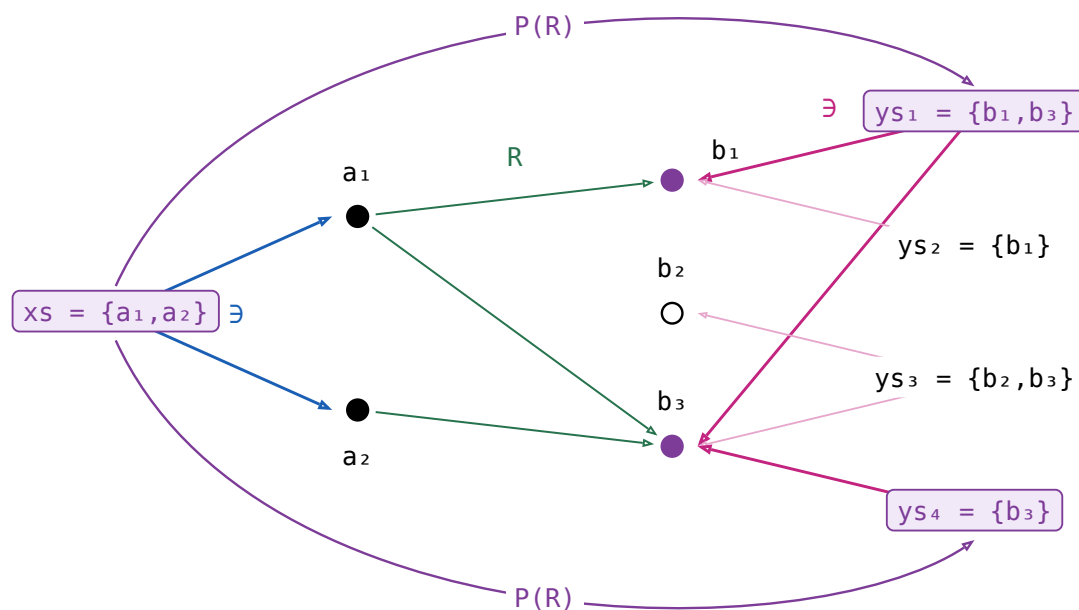


$$(\exists R)/\exists : E A \rightarrow E B \quad xs \ ((\exists R)/\exists) \ ys \iff \forall b \in ys. \exists a \in xs. a R b$$

$ys_3$  is rejected: it names  $b_2$ , which  $xs$  does not reach

The other half of the meet is that clause with the sets swapped and  $R$  turned round: every element of  $xs$  must reach  $ys$ :

(13f)



$$((\exists R^\circ)/\exists)^\circ : E A \rightarrow E B \quad xs \ ((\exists R^\circ)/\exists)^\circ \ ys \iff \forall a \in xs. \exists b \in ys. a R b$$

$ys_2$  is rejected too:  $a_2$ 's only image is  $b_3$ , which  $ys_2$  does not name

(13g)

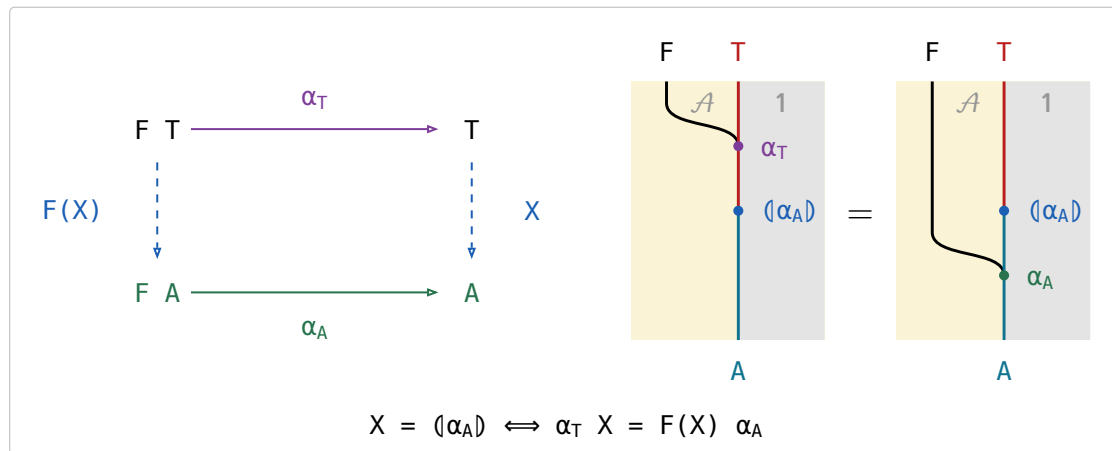
| law                                                                                                         | the reading                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $X \sqsubseteq P(R) \iff X \exists \sqsubseteq \exists R$ and $X^\circ \exists \sqsubseteq \exists R^\circ$ | One containment, and the same one at $R^\circ$ — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$ , and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$ .                                                                                                                              |
| $P(1) = \frac{\exists}{\exists} = 1$                                                                        | The straightness axiom verbatim: extensionality is $P$ 's unit law.                                                                                                                                                                                                                                                        |
| $P(f) = \frac{\exists f}{\exists}$ , for $f$ a map                                                          | In $\mathbf{Rel}$ , $xs \ P(f) \ ys \iff ys = \{f\ a \mid a \in xs\}$ . The half at $f^\circ$ says every $a \in xs$ has its $f\ a$ on $ys$ ; $f$ has just the one image per $a$ , so that already says $ys$ contains everything $xs$ reaches, which is the fraction's second half. For a map the two definitions coincide. |
| $P(R\ S) = P(R)\ P(S)$                                                                                      | $\sqsubseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(R\ S)$ .                                                                                                                                                              |

## §14. Catamorphism

let  $F$  be a relator and has **initial algebra**  $\alpha_T : F T \rightarrow T$  in the subcategory of functions.  $\alpha_T$  is also initial in the allegory:

(14a)

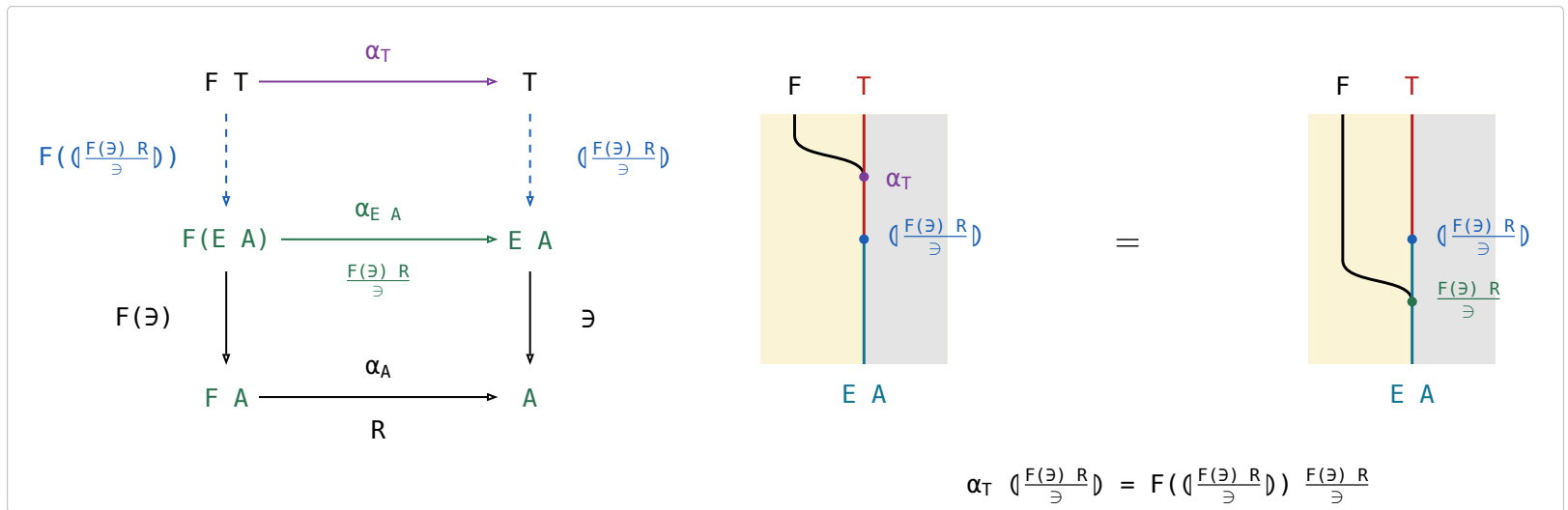
### §14.1. The defining equation



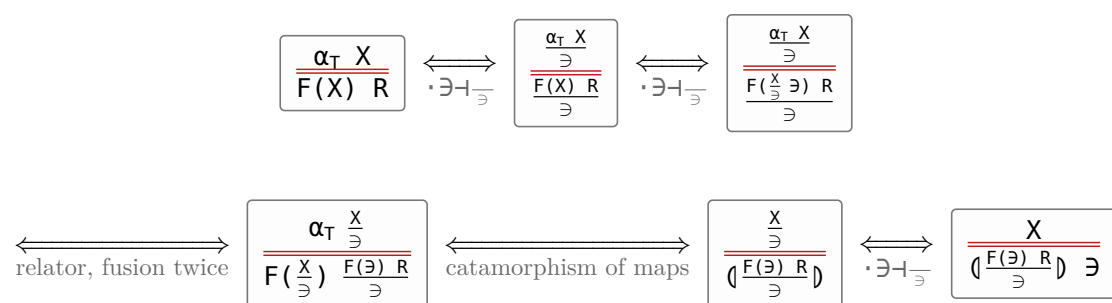
(14.1a)

An  $F$ -algebra is a **weakened** natural transformation. A transformation  $F \Rightarrow 1$  on  $\mathcal{A}$  would need a component  $F X \rightarrow X$  at every object and a commuting square at every arrow, but  $F$ -Algebra only need it works on  $T$  and  $B$ .

### §14.2. $(R) = (\frac{F(\exists) R}{\exists}) \exists$

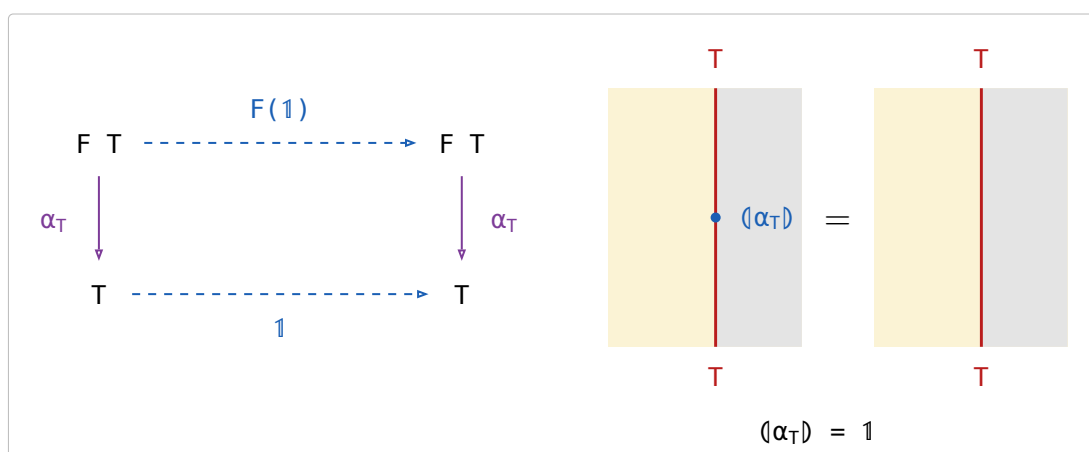


(14.2a)



(14.2b)

### §14.3. Reflection

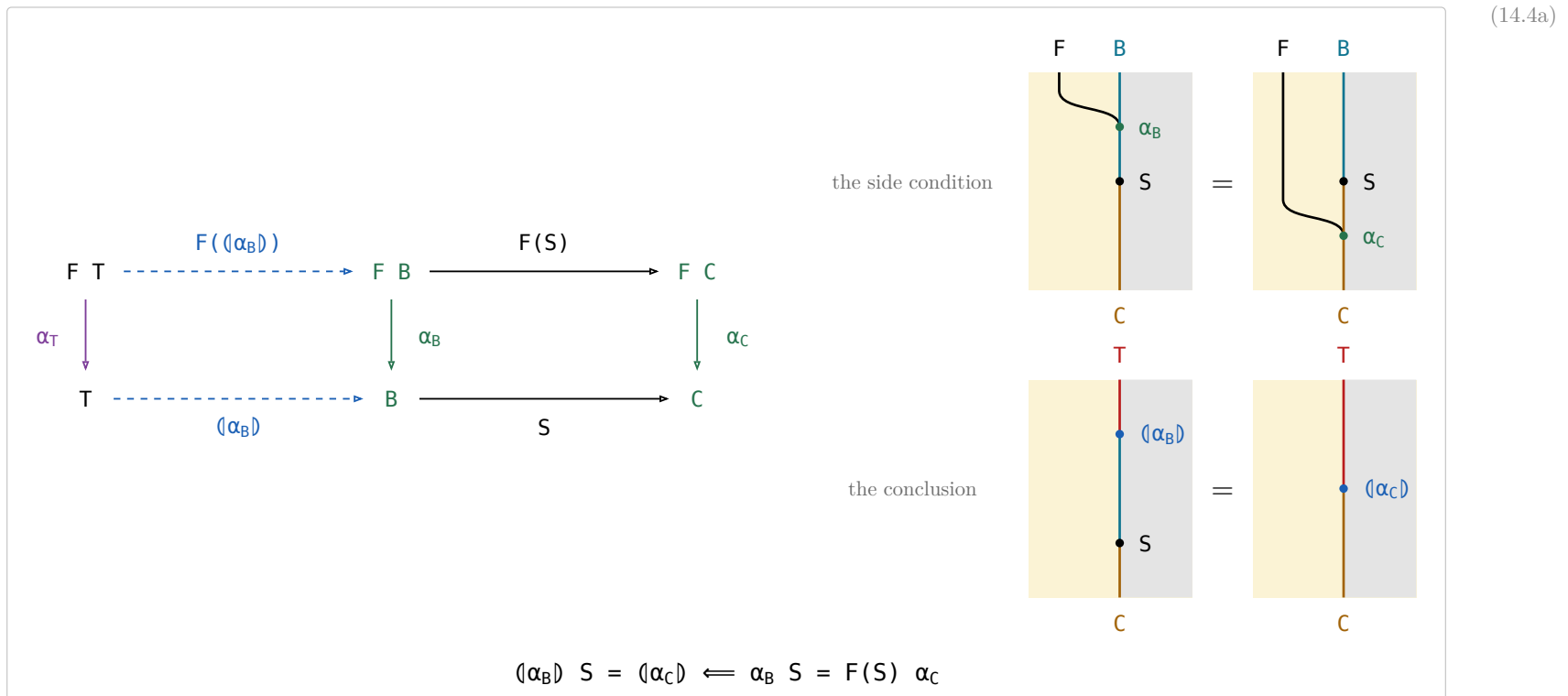


(14.3a)

Taking a value apart with  $\alpha_T$  and putting it straight back is doing nothing.

## §14.4. Fusion

Fusion rewrites  $(\alpha_B) S$  through a second algebra  $\alpha_C : F C \rightarrow C$  along an arrow  $S : B \rightarrow C$ .



The left square is  $(\alpha_B)$ 's own defining square and the right one is the side condition, so the outer rectangle says  $(\alpha_B) S$  satisfies the defining equation of  $(\alpha_C)$  — and uniqueness finishes it. A fold followed by  $S$  has collapsed into a single fold, which is how an intermediate structure is got rid of.

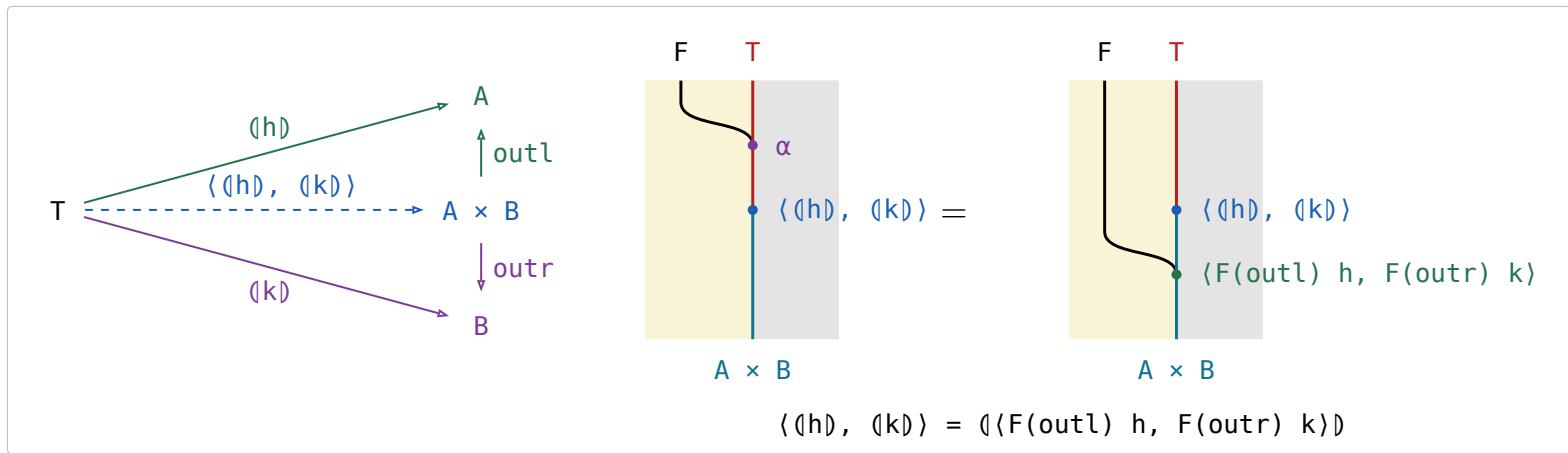
The side condition is (14.1a)'s picture again under  $\alpha_T \mapsto \alpha_B$ ,  $(\alpha_B) \mapsto S$ ,  $\alpha_B \mapsto \alpha_C$ : both say that the bead between the two merges is a homomorphism of  $F$ -algebras.

## §14.5. Catamorphism in $\mathcal{S}et$

(14.5a)

| name                 | definition                                                                                                                                                   | note                                                                                                                                   |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>listr</code>   | <code>listr A ::= nil   cons (A, listr A)</code>                                                                                                             | The cons-lists over A, the datatype every row below folds (B&dM p. 55).                                                                |
| <code>sum</code>     | <code>([zero, plus])</code>                                                                                                                                  | <code>plus(a, b) = a + b</code> .                                                                                                      |
| <code>length</code>  | <code>([zero, outr succ])</code>                                                                                                                             | <code>outr</code> drops the head and keeps the count of the tail, <code>succ</code> adds one for the head.                             |
| <code>average</code> | <code>(sum, length) div</code>                                                                                                                               | <code>div(m, n) = m/n</code> , with <code>div(0, 0) = 0</code> so <code>average</code> is total. Traverses the list twice.             |
| banana-split law     | <code>(\langle h \rangle, \langle k \rangle) = (\langle F(outl) h, F(outr) k \rangle)</code>                                                                 | Any fork of folds is a single fold, hence one traversal — $F$ the base functor.                                                        |
| what it reduces to   | $\alpha \langle \langle h \rangle, \langle k \rangle \rangle = F(\langle \langle h \rangle, \langle k \rangle \rangle) \langle F(outl) h, F(outr) k \rangle$ | All that (14.1a) leaves to check: the fork satisfies the defining equation, and uniqueness finishes it.                                |
| the instance         | <code>(sum, length) = ([zeros, pluss])</code>                                                                                                                | <code>zeros = (zero, zero)</code> and <code>pluss(a, (b, n)) = (a + b, n + 1)</code> , so <code>average</code> runs in one pass.       |
| <code>preds</code>   | <code>preds : Nat -&gt; [Nat], preds n = [n, n-1, ..., 1]</code>                                                                                             | B&dM Ex 3.6 (p. 57), posed and not answered there: apply Ex 3.4 to write <code>preds</code> as <code>(\langle k \rangle outl)</code> . |

(14.5b)



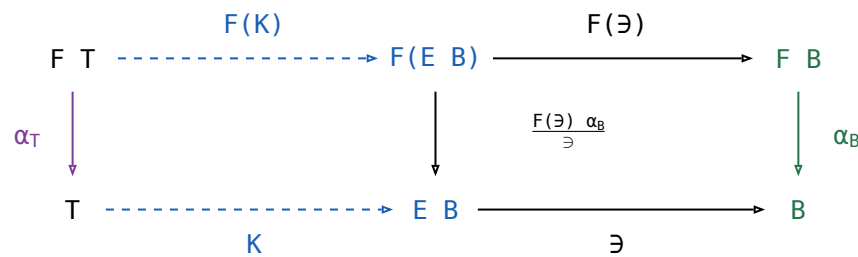
### The other laws.

The two  $\sqsubseteq$  **fusion** rows rewrite  $\langle \alpha_B \rangle S$  through the same  $\alpha_C$  and  $S$ . They are the only rows here that need the allegory **locally complete** — every hom-set a complete lattice — because they come from a least-fixed-point argument; the rest, the equality fusion (14.4a) included, needs only the initial algebra and the defining equation.

| name             | law                                                                                                         | what it says                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Lambek           | $\alpha_T^\circ = \alpha_T^{-1}$ , the initial algebra is an isomorphism                                    | A constructor can be undone — $\alpha_T^\circ$ takes a value apart into the parts it was built from.                                             |
| Eilenberg–Wright | $\langle \alpha_B \rangle \sqsubseteq \langle \frac{F(\exists) \alpha_B}{\exists} \rangle$                  | A relational fold is a deterministic fold of SETS: $\sqsubseteq$ pushes the nondeterminism into the power-object, where the fold is a map again. |
| Eilenberg–Wright | $\langle \alpha_B \rangle = \langle \frac{F(\exists) \alpha_B}{\exists} \rangle \sqsupset$                  | The same fact read back — fold deterministically into a set, then take a member of it.                                                           |
| fusion           | $\langle \alpha_C \rangle \sqsubseteq \langle \alpha_B \rangle S \iff F(S) \alpha_C \sqsubseteq \alpha_B S$ | Half of fusion: an inclusion between the two algebras is inherited by the folds.                                                                 |
| fusion           | $\langle \alpha_B \rangle S \sqsubseteq \langle \alpha_C \rangle \iff \alpha_B S \sqsubseteq F(S) \alpha_C$ | The other half, with both inclusions turned around.                                                                                              |

The two  $\sqsubseteq$  rows, drawn:

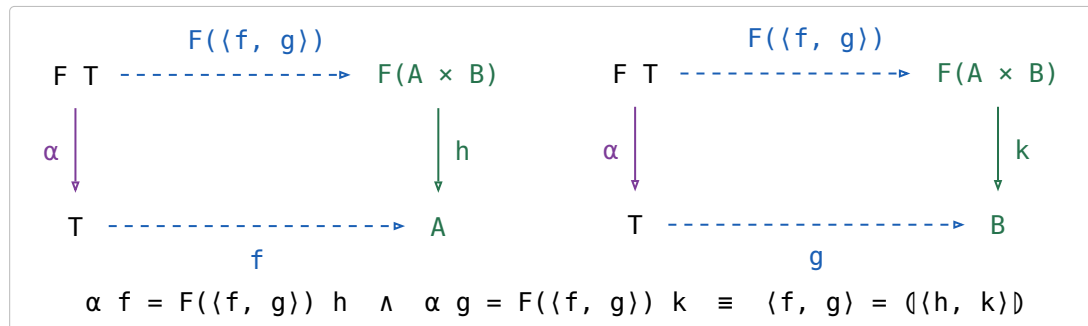
(14.5d)



The left square is that catamorphism's own defining square,  $K = \langle \frac{F(\exists) \alpha_B}{\exists} \rangle$ , and the right one is  $\sqsubseteq$ 's cancellation, so the outer rectangle says  $K \sqsubseteq$  satisfies the defining equation of  $\langle \alpha_B \rangle$  — and uniqueness finishes it.

### §14.5.1. Fokkinga's mutual recursion theorem

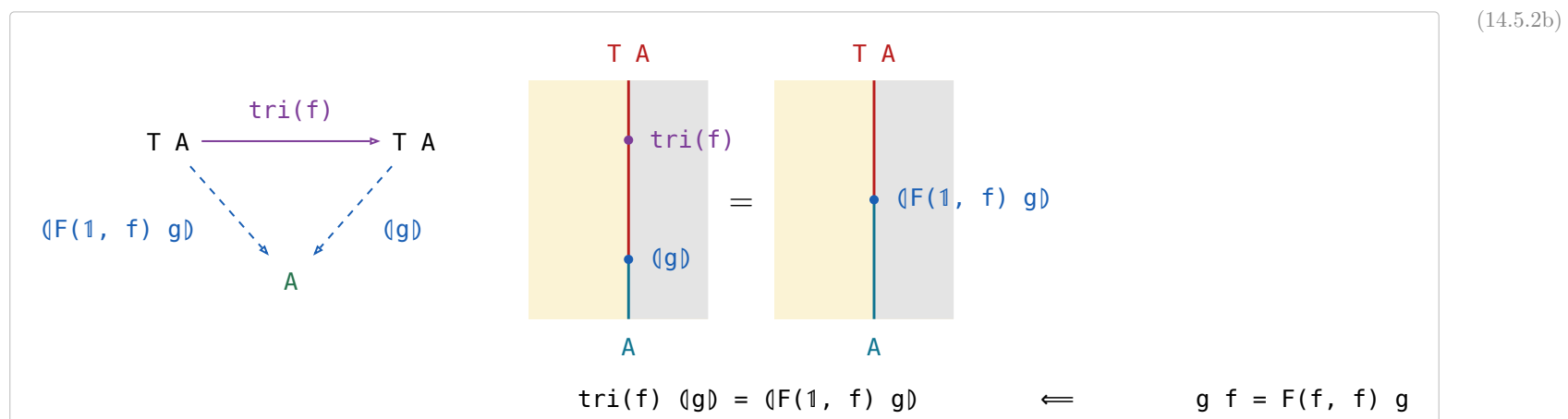
- $F\text{-Alg}(\mathcal{A})$  — the algebras and their homomorphisms — has binary products whenever  $\mathcal{A}$  does, and the forgetful  $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$  creates them: the product of  $h : F A \rightarrow A$  and  $k : F B \rightarrow B$  is carried by  $A \times B$ , with structure  $\langle F(\text{outl}) h, F(\text{outr}) k \rangle : F(A \times B) \rightarrow A \times B$ .
- $\text{outl}$  and  $\text{outr}$  are homomorphisms out of it, and  $\langle p, q \rangle : X \rightarrow A \times B$  is a homomorphism **iff**  $p$  and  $q$  both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (14.5a) IS that product's universal property read at the initial algebra:  $\langle h \rangle$  and  $\langle k \rangle$  are the unique homomorphisms to  $(A, h)$  and  $(B, k)$ , so their fork is the unique homomorphism into the product, which is  $\langle F(\text{outl}) h, F(\text{outr}) k \rangle$ .
- Fokkinga's theorem is **strictly more general** and is not that product: in (14.5.1a) the two arrows leave  $F(A \times B)$ , not  $F A$  and  $F B$ , so each sees BOTH components. The product is the case where they factor as  $F(\text{outl}) h$  and  $F(\text{outr}) k$ ; B&dM's Ex 3.4 is the other special case (p. 58). What it says: **any algebra on a product carrier is folded by a fork**.



### §14.5.2. Ruby triangles

| stage                       | definition                                                                                                                               | (14.5.2a) |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| informally (p. 58)          | $\text{tri}(f) [a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f a_1, \dots, f^i a_i, \dots, f^n a_n]$                                        |           |
| for cons-lists (p. 59)      | $\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{cons}] \rangle$                                                   |           |
| with the base functor named | $F(A, B) = 1 + A \times B, \alpha = [\text{nil}, \text{cons}], \text{ so } \text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$ |           |
| abstractly (p. 59)          | $F$ a bifunctor with initial type $(\alpha, T)$ : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$                                    |           |

For the definition to make sense  $f : A \rightarrow A$  is required, and then  $\text{tri}(f) : T A \rightarrow T A$ .



(14.5.2b) is Horner's rule, generalised from cons-lists to any initial type; B&dM call it that because for cons-lists it is the schoolbook method for evaluating a polynomial (p. 58). Fusion reduces it to  $F(1, T(f)) \alpha \langle g \rangle = F(1, \langle g \rangle) F(1, f) g$  (p. 59).

## §14.5.3. Depth of a tree

|                  |                                                                                                                                                                                                                                       |           |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| the datatype     | $\text{tree } A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$ , base functor $F(A, B) = A + B \times B$ ,<br>initial type $([\text{tip}, \text{bin}], \text{tree})$                                            | (14.5.3a) |
| the fold         | $\langle [g, h] \rangle$ is the unique $f$ with $f(\text{tip } a) = g\ a$ and $f(\text{bin } (x, y)) = h\ (f\ x, f\ y)$                                                                                                               |           |
| $\text{tree}(f)$ | $\text{tree}(f) = \langle F(f, 1) \rangle [\text{tip}, \text{bin}]$ ; pointwise $\text{tree}(f)(\text{tip } a) = \text{tip } (f\ a)$ and<br>$\text{tree}(f)(\text{bin } (x, y)) = \text{bin } (\text{tree}(f)\ x, \text{tree}(f)\ y)$ |           |
| $\text{max}$     | $\text{max} = \langle [1, \text{bmax}] \rangle$ , where $\text{bmax } (a, b)$ is the larger of $a$ and $b$                                                                                                                            |           |
| $\text{depths}$  | $\text{depths} = \text{tree}(\text{zero})\ \text{tri}(\text{succ})$ — replaces every tip by its depth in the tree, $\text{zero}$ the constant<br>function returning 0 and $\text{succ}$ the successor                                 |           |
| $\text{depth}$   | $\text{depth} = \text{depths}\ \text{max}$                                                                                                                                                                                            |           |

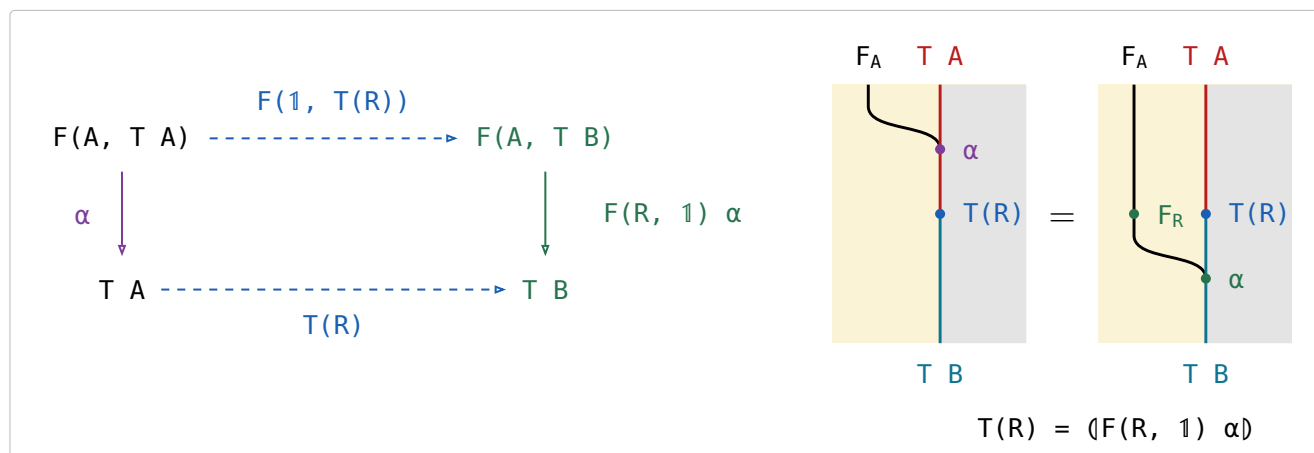
## §15. Type functor

$F$  a **binary** relator:  $F(R, S)$  is its action on a pair, and  $F(X)$  abbreviates  $F(1, X)$ , the  $F$  of the catamorphism section. For every object  $A$  the initial algebra is  $\alpha : F(A, T A) \rightarrow T A$ , among the maps. The **type functor**  $T$  acts on an arrow  $R : A \rightarrow B$  by

$$T(R) = (F(R, 1) \alpha) : T A \rightarrow T B$$

$F$  is the **base functor** — one layer of the structure, acting on the recursive position — while  $T$  is the datatype itself, acting on the parameter. For cons-lists,  $\text{list}(R) = ([\text{nil}, (R \otimes 1) \text{ cons}])$ , which is  $\text{map}(R)$ .

A bifunctor puts its two arguments in two different places. The wire is the partial application  $F_A = F(A, -)$ , and an arrow in the SECOND argument is a bead on the object wire with the  $F$  wire running past it — the catamorphism section's rule, unchanged — while an arrow  $R : A \rightarrow B$  in the FIRST induces a natural transformation  $F_R : F_A \Rightarrow F_B$ , which is a bead ON the  $F$  wire. Marsden draws bifunctors exactly so ([CatString.pdf](#), **Bifunctors**): side-by-side is already spent on functor composition, so one argument has to go into the wire's name.



$T(R)$  is the unique arrow making it commute; there is no  $\sqsubseteq$  in it.

Two beads at one height on two different wires are one action, here  $F(R, T(R))$ , so the right panel reads  $F(R, T(R)) \alpha$  and the left one  $\alpha T(R)$ . Their relative height says nothing: slide  $F_R$  below the  $T(R)$  bead and the panel reads  $F(1, T(R)) F(R, 1)$  and then  $\alpha$ , which is the square's top row followed by its right vertical; slide it above and the panel reads  $F(R, 1) F(1, T(R))$ , the route through  $F(B, T A)$  the square leaves out. Identifying those two is the exchange condition every bifunctor satisfies (Hinze & Marsden, Ex 1.23), and in this calculus there is nothing to identify.

| name                   | law                                 | what it says                                                                                                                           |
|------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| the defining equation  | $T(R) = (F(R, 1) \alpha)$           | Rebuild the structure with $\alpha$ , applying $R$ to the parameter on the way; that is $\text{map}(R)$ .                              |
| functor                | $T(1) = 1$ and $T(R) T(S) = T(R S)$ | Mapping the identity changes nothing, and two maps in a row are one map.                                                               |
| type functor fusion    | $T(R) (Q) = (F(R, 1) Q)$            | A map followed by a fold is a single fold — the mapped structure is never built.                                                       |
| naturality of $\alpha$ | $\alpha T(R) = F(R, T(R)) \alpha$   | Building and then mapping is the same as mapping the parts and then building, so $\alpha$ is natural from $G(R) = F(R, T(R))$ to $T$ . |
| type relator           | $T(R)^\circ = T(R^\circ)$           | A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.                                   |

Type functor fusion is the equality fusion (14.4a) applied to  $T(R)$ 's own defining algebra, whose side condition holds because  $F$  is a bifunctor —  $F(R, 1) F(1, (Q)) = F(R, (Q)) = F(1, (Q)) F(R, 1)$  — so it too needs only the initial algebra and the defining equation.



(15d)

$$\begin{array}{ccc}
 F(A, T A) & \xrightarrow{F(R, T(R))} & F(B, T B) \\
 \downarrow \alpha & & \downarrow \alpha \\
 T A & \xrightarrow{T(R)} & T B
 \end{array}$$

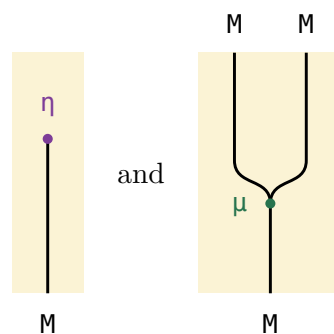
It commutes strictly, and it is (15b) with the right vertical's two halves separated: push  $F(R, 1)$  up into the top row, where it meets  $F(1, T(R))$  as the one action  $F(R, T(R))$ , and  $\alpha$  is left as the vertical. In the string diagram that is the  $F_R$  bead sliding up past  $T(R)$ , and what stays put is  $\alpha$ , one bead falling past  $T(R)$  from the row above to the row below — which is what “ $\alpha$  is natural” means.

## §16. Monad

An endofunctor  $M : \mathcal{A} \rightarrow \mathcal{A}$  with two natural transformations, the **unit**  $\eta : \text{Id} \Rightarrow M$  and the **multiplication**  $\mu : M \circ M \Rightarrow M$ , subject to a unit law and an associativity law:

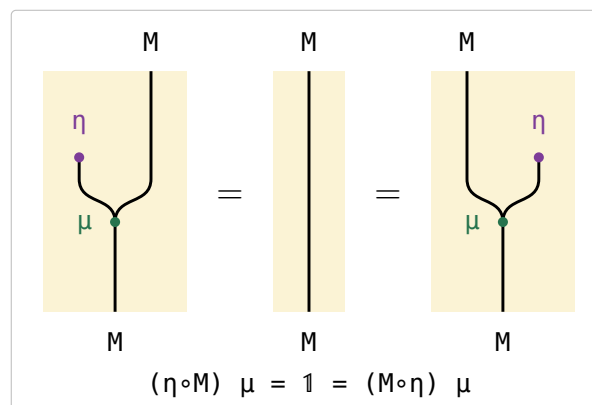
$$(\eta \circ M) \mu = 1 = (M \circ \eta) \mu \qquad (\mu \circ M) \mu = (M \circ \mu) \mu$$

(16a)



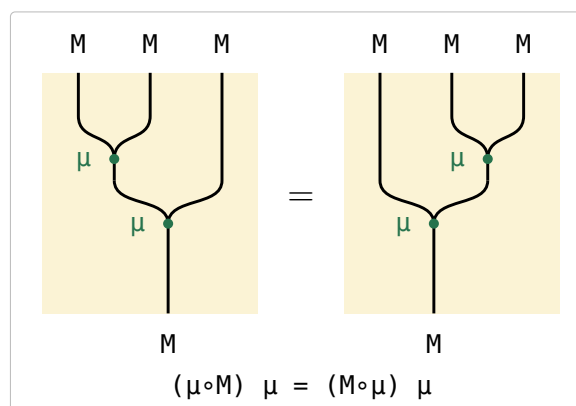
(16b)

In (16b) the identity functor is a REGION and not a wire, so  $\eta : \text{Id} \Rightarrow M$  has nothing above it and the  $M$  wire simply begins at its bead — the book's lollipop.  $\mu : M \circ M \Rightarrow M$  is the two  $M$  wires of  $M \circ M$  merging into one, its tuning fork.



(16c)

The lollipop is absorbed into the fork from either side and a bare wire is left, which is what  $1$  is in this calculus: nothing is drawn for it.



(16d)

The two nestings of the fork agree, so three  $M$ s collapse to one whichever pair is multiplied first.

### §16.1. Monad in $\mathcal{S}et$

| monad                              | $\eta$              | $\mu$                           | $\eta$ on elements              | $\mu$ on elements                                          |
|------------------------------------|---------------------|---------------------------------|---------------------------------|------------------------------------------------------------|
| $\text{Id } A = A$                 | $1$                 | $1$                             | $\eta A a = a$                  | $\mu A a = a$                                              |
| $\text{Exc } A = A + E$            | $\iota_l$           | $1 \nabla \iota_r$              | $\eta A a = \iota_l a$          | $\mu A (\iota_l y) = y$<br>$\mu A (\iota_r e) = \iota_r e$ |
| $\text{State } A = (A \times P)^P$ | $\text{curry } 1$   | $(\text{apply}_{A \times P})^P$ | $\eta A a = \lambda x . (a, x)$ | $\mu A f = \lambda x . g y \text{ where } (g, y) = f x$    |
| $\text{Pow } A = E A$              | $\frac{1}{\exists}$ | $\cup$                          | $\eta A a = \{a\}$              | $\mu A Xs = \cup Xs$                                       |

(16.1a)

## §16.2. The state monad

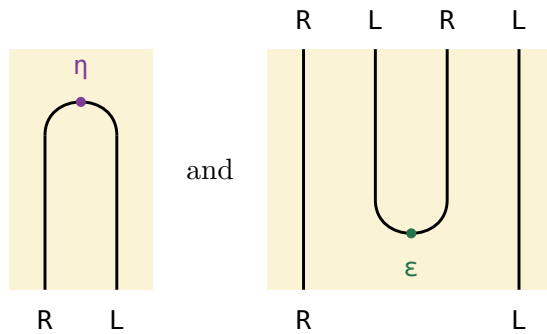
For a fixed set  $P$  of states, IntroString p. 67 defines

$$\text{State } A = (A \times P)^P \quad \eta A = \text{curry } (1 : A \times P \rightarrow A \times P) \quad \mu A = (\text{apply}_{A \times P})^P$$

Both arrows come from the curry adjunction  $L \dashv R$ ,  $L A = A \times P$  and  $R B = B^P$  (IntroString p. 96):

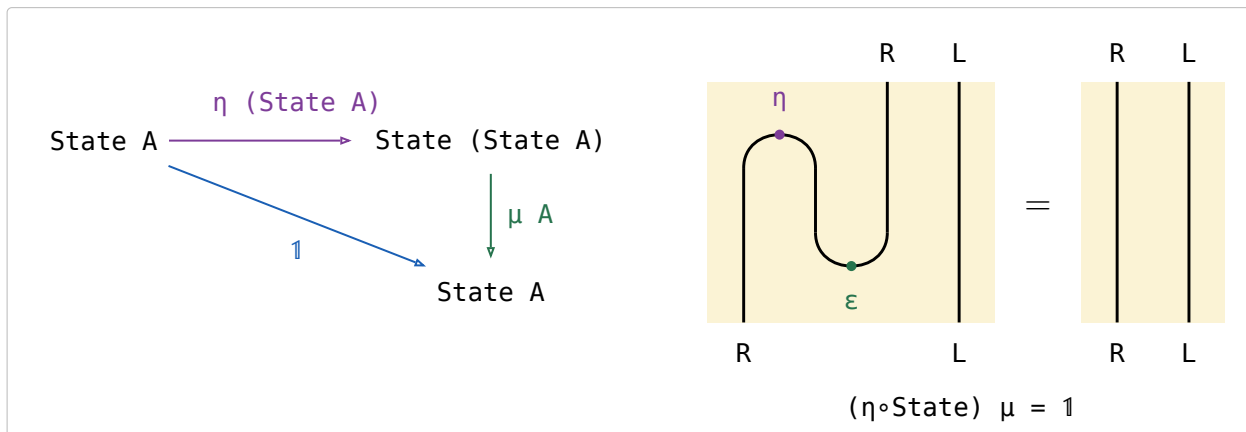
- $\text{State} = R \circ L$ .
- $\eta$  is the unit  $\text{Id} \Rightarrow R \circ L$ , which is why it is the curried identity.
- the counit  $\epsilon : L \circ R \Rightarrow \text{Id}$  is **apply**.
- $\mu = R \circ \epsilon \circ L$ , that counit with a functor on each side: in  $\mu A = (\text{apply}_{A \times P})^P$  the subscript  $A \times P$  is  $L A$ , the functor INSIDE, and the outer  $(-)^P$  is  $R$ , the functor OUTSIDE.
- $\dashv$  composes nothing; it only names the left adjoint first.

Applicative order stands  $R$  left and  $L$  right in  $\text{State} = R \circ L$ . The identity functor is a region, so a unit  $\text{Id} \Rightarrow R \circ L$  has nothing above it and two wires below — one strand that **OPENS** the pair; a counit  $L \circ R \Rightarrow \text{Id}$  is that strand upside down, **CLOSING** a pair that arrives from above.



(16.2a)

In (16.2a),  $\text{State} \circ \text{State}$  is the four wires  $R L R L$ , and  $\mu = R \circ \epsilon \circ L$  closes the middle pair — the  $L \circ R$  — leaving the outer  $R$  and  $L$ , which is  $\text{State}$  again. The two wires that run past the turn untouched are the two functors composed on,  $R$  outside and  $L$  inside.



(16.2b)

(16.2b) is the zig-zag.  $\eta \circ \text{State}$  opens a new  $R L$  left of the pair already there and  $\mu$  closes the middle two — the new  $L$  against the old  $R$  — leaving the new  $R$  and the old  $L$ , which is  $\text{State}$  unchanged. Strip the old  $L$ , which never moves, and what zig-zags is the curry adjunction's snake equation  $(\eta \circ R)(R \circ \epsilon) = 1$  (IntroString p. 93): (16c) read at  $\text{State}$  is that triangle identity with  $L$  composed on the inside.

In Set it is beta-reduction.  $\eta (\text{State } A) f = \lambda x . (f, x)$  pairs the computation with the state, and  $\mu A$  applies it back,  $\lambda x . f x$ , which is  $f$ .

## §17. Combinatorial functions

(17a)

| name         | definition                                                                                                                             | note                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| list         | $\text{list } A ::= \text{nil} \mid \text{cons } (A, \text{list } A)$<br>$\text{list}(R) = ([\text{nil}, (R \otimes 1) \text{ cons}])$ | B&dM's <code>listr</code> under the short name it keeps from p. 125 on.                                                                                                                                                                                                                                                                                                                                                                  |
| subseq       | $([\text{nil}, \text{cons } u \text{ outr}])$                                                                                          | $xs \text{ subseq } ys$ : $ys$ is $xs$ with elements dropped — <code>cons</code> keeps the head, <code>outr</code> drops it.                                                                                                                                                                                                                                                                                                             |
| prefix       | $([\text{nil}, \text{nil } u \text{ cons}]) = \text{cat}^\circ \text{ outl} = \text{init}^*$                                           | $ys$ is an initial segment of $xs$ ; the first <code>nil</code> is where it stops early. $\text{init} \triangleq \text{snoc}^\circ \text{ outl}$ .                                                                                                                                                                                                                                                                                       |
| suffix       | $\text{cat}^\circ \text{ outr} = \text{tail}^*$                                                                                        | The dual, $\text{tail} \triangleq \text{cons}^\circ \text{ outr}$ ; as a catamorphism it needs <code>snoc</code> -lists.                                                                                                                                                                                                                                                                                                                 |
| segment      | <code>suffix prefix</code>                                                                                                             | A contiguous stretch of $xs$ : a suffix, then a prefix of that.                                                                                                                                                                                                                                                                                                                                                                          |
| partition    | $\text{concat}^\circ, \text{concat} \triangleq ([\text{nil}, \text{cat}])$                                                             | <code>cat</code> restricted to a non-empty first argument, so $ys$ is a list of non-empty segments of $xs$ .                                                                                                                                                                                                                                                                                                                             |
| inits, tails | implementations of $\frac{\text{prefix}}{\ni}$ and $\frac{\text{suffix}}{\ni}$                                                         | <code>inits</code> by increasing length, <code>tails</code> by decreasing — opposite orders, which is what <code>splits</code> $\triangleq \langle \text{inits}, \text{tails} \rangle$ <code>zip</code> needs to implement $\frac{\text{cat}^\circ}{\ni}$ .                                                                                                                                                                              |
| filter(p)    | $\frac{\text{subseq } \text{list}(p)}{\ni} \max R, R \triangleq \text{length} \leq \text{length}^\circ$                                | The longest subsequence of $xs$ whose every element passes $p$ . $p$ is a <b>coreflexive</b> and <code>list(p)</code> the relator applied to it, so <code>list(p)</code> is the test “every element passes” — no boolean anywhere. The longest is unique: only one subsequence keeps exactly the passing elements. The program is $([\text{nil}, (\text{outl } p \rightarrow \text{cons}, \text{outr})])$ . Ex 7.41; $\max R$ is (18.1a) |
| takewhile(p) | $\frac{\text{prefix } \text{list}(p)}{\ni} \max R$                                                                                     | The same with <code>prefix</code> for <code>subseq</code> : the longest prefix whose every element passes $p$ . Unique, since the prefixes of one list are linearly ordered by length. Ex 7.39                                                                                                                                                                                                                                           |
| mss          | $\frac{\text{segment } \text{sum}}{\ni} \max \leq$                                                                                     | Maximum segment sum. <code>segment</code> = <code>suffix prefix</code> splits it into $\frac{\text{prefix } \text{sum}}{\ni} \max \leq$ on each suffix, which the greedy theorem turns into a catamorphism. Ex 7.40                                                                                                                                                                                                                      |

## §18. Optimisation Problems

### §18.1. Minimum and maximum

(18.1a)

For  $R : A \rightarrow A$ ,  $\min R \triangleq \exists n (\in R^\circ) : E A \rightarrow A$ ,  $\max R \triangleq \min R^\circ$ .

$xs (\min R) x \iff x \in xs \wedge (\forall y \in xs. x R y)$

$xs (\min R) x \iff (x \text{ in } xs) \text{ and all } x R \setminus xs \text{ in } q$

$\min R = \exists n \text{ all } R^\circ \text{ all } R \triangleq \in R, q's \text{ all}; \text{ the chains below keep it written } \in \setminus$

(18.1b)

| the law                                                                                                                     | what it says                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| $X \sqsubseteq \min R \iff X \sqsubseteq \exists \text{ and } X^\circ \exists \sqsubseteq R$                                | in the set, and below every element of it                                                                     |
| $\frac{1}{\exists} (\in R) = R$                                                                                             | bounding a singleton is bounding its element                                                                  |
| $\frac{S}{\exists} (\in R) = S^\circ \setminus R$                                                                           | bound $S$ 's image without building the set                                                                   |
| $\text{union } (\in R) = \in (\in R)$<br>$\text{union} \triangleq \frac{\exists \exists}{\exists} : E(E A) \rightarrow E A$ | bound a union by bounding each member set                                                                     |
| $\frac{1}{\exists} \min R = 1 n R$                                                                                          | a singleton's minimum is its element, where $R$ is reflexive<br>$\frac{S}{\exists} \min R \text{ at } S := 1$ |
| $\frac{S}{\exists} \min R = S n (S^\circ \setminus R^\circ)$                                                                | an $S$ -value that points to every $S$ -value                                                                 |
| $\frac{S}{\exists} \min R = \frac{S}{\exists} \min(R n S^\circ S)$                                                          | only $R$ between values $S$ gives one argument counts — context                                               |
| $E(S) \min R = (\exists S) n ((\exists S)^\circ \setminus R^\circ)$                                                         | the same for the image of a set<br>$\frac{S}{\exists} \min R \text{ at } S := \exists S$                      |
| $P(f) \min R = \min(f R f^\circ) f$                                                                                         | shunt a function through a minimum                                                                            |
| $P(S) \min R = (\exists S) n (\in (S R^\circ))$<br>$R \text{ reflexive}$                                                    | fusion with the power relator<br>$\exists$ is the only proof here that tabulates                              |
| $P(S) \min R \sqsubseteq (\exists S) n (\in (S R^\circ))$                                                                   | the half of the row above that costs nothing                                                                  |
| $P(\min R) \min R \sqsubseteq \text{union } \min R$<br>$R \text{ a preorder}$                                               | a minimum in each set, then a minimum of those                                                                |
| $P(\min R) \min R = P(\text{Dom } (\min R)) \text{ union } \min R$<br>$R \text{ a preorder}$                                | the same as an equality, once empty sets are dropped                                                          |

#### §18.1.1. $X \sqsubseteq \min R \iff X \sqsubseteq \exists \text{ and } X^\circ \exists \sqsubseteq R$

(18.1.1a)

$$\boxed{\frac{X}{\min R}} \xLeftrightarrow{\Delta \dashv n} \boxed{\frac{X}{\exists} \quad \frac{X}{\in R^\circ}} \xLeftrightarrow{T \cdot \dashv T \setminus} \boxed{\frac{X}{\exists} \quad \frac{X}{\in R^\circ}} \xLeftrightarrow{\circ} \boxed{\frac{X}{\exists} \quad \frac{X^\circ \exists}{R}}$$

#### §18.1.2. $\frac{1}{\exists} (\in R) = R$

(18.1.2a)

$$\boxed{\frac{X}{\frac{1}{\exists} (\in R)}} \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\frac{1^\circ X}{\frac{1}{\exists} \in R}} \xLeftrightarrow{T \cdot \dashv T \setminus} \boxed{\frac{\in \frac{1^\circ X}{\frac{1}{\exists}}}{R}} \xLeftrightarrow{\circ} \boxed{\frac{(\frac{1}{\exists} \exists)^\circ X}{R}} \xLeftrightarrow{i \dashv E} \boxed{\frac{X}{R}}$$

#### §18.1.3. $\frac{S}{\exists} (\in R) = S^\circ \setminus R$

(18.1.3a)

$$\boxed{\frac{X}{\frac{S}{\exists} (\in R)}} \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\frac{S^\circ X}{\frac{S}{\exists} \in R}} \xLeftrightarrow{T \cdot \dashv T \setminus} \boxed{\frac{\in \frac{S^\circ X}{\frac{S}{\exists}}}{R}} \xLeftrightarrow{\circ} \boxed{\frac{(\frac{S}{\exists} \exists)^\circ X}{R}} \xLeftrightarrow{\cdot \exists \dashv \frac{S}{\exists}} \boxed{\frac{S^\circ X}{R}} \xLeftrightarrow{T \cdot \dashv T \setminus} \boxed{\frac{X}{S^\circ \setminus R}}$$

#### §18.1.4. $\text{union } (\in R) = \in (\in R)$

(18.1.4a)

$$\boxed{\frac{X}{\text{union } (\in R)}} \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\frac{\text{union}^\circ X}{\in R}} \xLeftrightarrow{T \cdot \dashv T \setminus} \boxed{\frac{\in \text{union}^\circ X}{R}} \xLeftrightarrow{\circ} \boxed{\frac{(\text{union } \exists)^\circ X}{R}} \xLeftrightarrow{\cdot \exists \dashv \frac{S}{\exists}} \boxed{\frac{(\exists \exists)^\circ X}{R}} \xLeftrightarrow{\circ, T \cdot \dashv T \setminus} \boxed{\frac{X}{\in (\in R)}}$$

§18.1.5.  $\frac{S}{\exists} \min R = S \cap (S^\circ \setminus R^\circ)$

(18.1.5a)

$$\begin{array}{c}
 \boxed{\frac{X}{\frac{S}{\exists} \min R}} \xleftrightarrow{f^\circ \cdot \neg f \cdot} \boxed{\frac{\frac{S^\circ}{\exists} X}{\min R}} \xleftrightarrow{\Delta \neg \cap} \boxed{\frac{\frac{S^\circ}{\exists} X}{\exists}} \xleftrightarrow{f^\circ \cdot \neg f \cdot} \boxed{\frac{X}{\frac{S}{\exists} \exists}} \xleftrightarrow{\cdot \exists \neg \neg, (18.1.3a)} \boxed{\frac{X}{S}} \xleftrightarrow{\Delta \neg \cap} \boxed{\frac{X}{S \cap (S^\circ \setminus R^\circ)}}
 \end{array}$$

§18.1.6.  $\frac{S}{\exists} \min R = \frac{S}{\exists} \min(R \cap S^\circ S)$

(18.1.6a)

$$\begin{array}{c}
 \boxed{\frac{X}{\frac{S}{\exists} \min(R \cap S^\circ S)}} \xleftrightarrow{(18.1.5a), ^\circ} \boxed{\frac{X}{S \cap (S^\circ \setminus (R^\circ \cap S^\circ S))}} \xleftrightarrow{\Delta \neg \cap, T \cdot \neg T \setminus} \boxed{\frac{X}{S}} \xleftrightarrow{\Delta \neg \cap, S^\circ \cdot \text{monotone}} \boxed{\frac{X}{S}} \xleftrightarrow{T \cdot \neg T \setminus, \Delta \neg \cap} \boxed{\frac{X}{S \cap (S^\circ \setminus R^\circ)}} \xleftrightarrow{(18.1.5a)} \boxed{\frac{X}{\frac{S}{\exists} \min R}}
 \end{array}$$

§18.1.7.  $P(f) \min R = \min(f R f^\circ) f$

(18.1.7a)

$$\begin{array}{c}
 \boxed{\frac{P(f) \min R}{f \text{ a map}}} \xleftrightarrow{P = E \text{ on maps, (18.1.5a)}} \boxed{(\exists f) \cap ((\exists f)^\circ \setminus R^\circ)} \xleftrightarrow{^\circ, T \cdot \neg T \setminus, f^\circ \cdot \neg f \cdot} \boxed{(\exists f) \cap (E \setminus (f R^\circ))} \\
 \xleftrightarrow{\text{modular law, } f \text{ simple}} \boxed{(\exists \cap ((E \setminus (f R^\circ)) f^\circ)) f} \xleftrightarrow{\cdot f \neg \cdot f^\circ, ^\circ, \min} \boxed{\min(f R f^\circ) f}
 \end{array}$$

§18.1.8.  $P(S) \min R = (\exists S) \cap (E \setminus (S R^\circ))$

(18.1.8a)

$(p, q)$  tabulates  $W \triangleq (\exists S) \cap (E \setminus (S R^\circ))$ , and  $y \triangleq \frac{(p \exists S) \cap (q R)}{\exists}$ , a map.

(18.1.8b)

$$\boxed{\frac{W}{P(S) \min R}} \xleftrightarrow{p^\circ q = W, 1 \sqsubseteq y y^\circ} \boxed{\frac{p^\circ y}{P(S)}} \quad \boxed{\frac{y^\circ q}{\min R}}$$

(18.1.8c)

$$\boxed{\frac{y^\circ q}{\exists}} \xleftrightarrow{f^\circ \cdot \neg f \cdot, \cdot \exists \neg \neg} \boxed{\frac{q}{(p \exists S) \cap (q R)}} \xleftrightarrow{\Delta \neg \cap, f^\circ \cdot \neg f \cdot} \boxed{\frac{p^\circ q}{\exists S}} \quad \boxed{\frac{1}{R}} \text{ R reflexive}$$

(18.1.8d)

$$\begin{array}{c}
 \boxed{\frac{E y^\circ q}{R^\circ}} \xleftrightarrow{\cdot \exists \neg \neg, ^\circ, \text{meet}} \boxed{\frac{R^\circ q^\circ q}{R^\circ}} \text{ q simple} \\
 \boxed{\frac{p^\circ y}{P(S)}} \xleftrightarrow{\Delta \neg \cap, \cdot T \neg / T, ^\circ, f^\circ \cdot \neg f \cdot} \boxed{\frac{p^\circ y \exists}{\exists S}} \quad \boxed{\frac{p \exists}{y \exists S^\circ}} \\
 \boxed{\frac{p^\circ y \exists}{\exists S}} \xleftrightarrow{\cdot \exists \neg \neg, \text{meet}} \boxed{\frac{p^\circ p \exists S}{\exists S}} \text{ p simple} \\
 \boxed{\frac{p \exists}{y \exists S^\circ}} \xleftrightarrow{\text{modular law}} \boxed{\frac{p \exists}{(p \exists) \cap (q R S^\circ)}} \xleftrightarrow{1 \sqsubseteq q q^\circ} \boxed{\frac{q^\circ p \exists}{R S^\circ}} \xleftrightarrow{^\circ, T \cdot \neg T \setminus} \boxed{\frac{W}{E \setminus (S R^\circ)}} \text{ W's right half}
 \end{array}$$

### §18.1.9. $P(S) \min R \sqsubseteq (\exists S) \cap (\epsilon \setminus (S R^\circ))$

(18.1.9a)

$$\begin{array}{ccccc}
 \boxed{\frac{P(S) \min R}{(\exists S) \cap (\epsilon \setminus (S R^\circ))}} & \xleftarrow{\Delta \dashv \cap, T \cdot \dashv T \setminus} & \boxed{\frac{P(S) \min R}{\exists S}} & \xleftarrow{\text{UP of min}} & \boxed{\frac{P(S) \exists}{\exists S}} \\
 & & \boxed{\frac{\epsilon P(S) \min R}{S R^\circ}} & & \boxed{\frac{\epsilon P(S)}{S \epsilon}}
 \end{array}$$

### §18.1.10. $P(\min R) \min R \sqsubseteq \text{union } \min R$

(18.1.10a)

$$\begin{array}{ccccc}
 \boxed{\frac{P(\min R) \min R}{\text{union } \min R}} & \xleftrightarrow{(18.1.5a)} & \boxed{\frac{P(\min R) \min R}{(\exists \exists) \cap ((\exists \exists)^\circ \setminus R^\circ)}} & \xleftarrow{\circ, \Delta \dashv \cap, T \cdot \dashv T \setminus} & \boxed{\frac{P(\min R) \min R}{\exists \exists}} \\
 \text{R a preorder} & & & & \boxed{\frac{\epsilon \in P(\min R) \min R}{R^\circ}} \\
 & & & & \xleftarrow{\text{UP of min, R transitive}} \boxed{\frac{P(\min R) \exists}{\exists \min R}} \\
 & & & & \boxed{\frac{\epsilon P(\min R)}{\min R \epsilon}}
 \end{array}$$

### §18.1.11. $P(\min R) \min R = P(\text{Dom } (\min R)) \text{ union } \min R$

(18.1.11a)

$$\begin{array}{ccc}
 \boxed{P(\min R) \min R} & \xrightarrow{\text{Dom } (\min R) \min R = \min R} & \boxed{P(\text{Dom } (\min R) \min R) \min R} \\
 \xrightarrow{\text{P a relator}} & \boxed{P(\text{Dom } (\min R)) P(\min R) \min R} & \xrightarrow{(18.1.10a)} \boxed{P(\text{Dom } (\min R)) \text{ union } \min R}
 \end{array}$$

(18.1.11b)

$$\begin{array}{ccc}
 \boxed{\frac{P(\text{Dom } (\min R)) \text{ union } \min R}{P(\min R) \min R}} & \xleftarrow{(18.1.8b) \text{ at } S := \min R, \Delta \dashv \cap, T \cdot \dashv T \setminus} & \boxed{\frac{P(\text{Dom } (\min R)) \text{ union } \min R}{\exists \min R}} \\
 & & \boxed{\frac{\epsilon P(\text{Dom } (\min R)) \text{ union } \min R}{\min R R^\circ}}
 \end{array}$$

$$\begin{array}{ccc}
 \boxed{\frac{P(\text{Dom } (\min R)) \text{ union } \min R}{\exists \min R}} & \xleftarrow{P(\text{Dom } (\min R)) \sqsubseteq 1} & \boxed{\frac{\text{union } \min R}{\exists \min R}} \\
 \xleftarrow{T(U \cap V) \sqsubseteq T U \cap T V, \cdot \exists \dashv \frac{\cdot}{\exists}, (18.1.4a)} & \boxed{\frac{(\exists \exists) \cap (\epsilon \setminus (\epsilon \setminus R^\circ))}{\exists \min R}} & \xleftarrow{\text{modular law}} \boxed{\frac{\exists (\exists \cap (\epsilon \setminus (\epsilon \setminus R^\circ)))}{\exists (\exists \cap (\epsilon \setminus R^\circ))}} \\
 & & \text{counit of } T \cdot \dashv T \setminus
 \end{array}$$

$$\boxed{\frac{\epsilon P(\text{Dom } (\min R)) \text{ union } \min R}{\min R R^\circ}} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\frac{\text{Dom } (\min R) \epsilon \text{ union } \min R}{\min R R^\circ}}$$

$$\begin{array}{ccc}
 \xleftarrow{\text{Dom } (\min R) \sqsubseteq \min R (\min R)^\circ} & \boxed{\frac{\min R (\min R)^\circ \epsilon \text{ union } \min R}{\min R R^\circ}} & \xleftarrow{(\min R)^\circ \sqsubseteq \epsilon, \epsilon \in \text{union } \sqsubseteq \epsilon} \boxed{\frac{\min R \epsilon \min R}{\min R R^\circ}} \\
 & & \text{UP of min}
 \end{array}$$

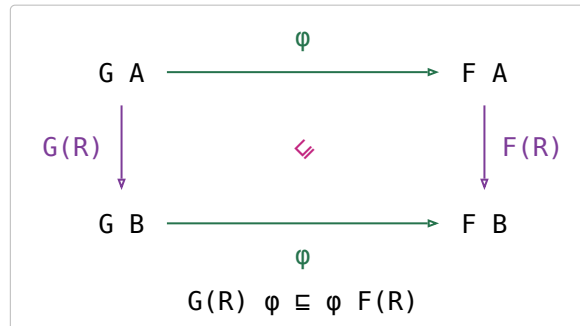
## §18.2. Lax natural transformations

$\varphi_A : G A \rightarrow F A$  is a **lax natural transformation**  $\varphi : G \Rightarrow F$  when, for every  $R : A \rightarrow B$ ,  $G(R) \varphi \sqsubseteq \varphi F(R)$ . (5.13)

(18.2a)

$\exists : P \Rightarrow \text{Id}$  — that is  $P(R) \exists \sqsubseteq \exists R$ , (13g)'s first row at  $X := P(R)$ .

It is enough to check **maps**, and there it is an equality  $G(f) \varphi = \varphi F(f)$ : laxness is about relations only. Theorem 5.2



(18.2b)

## §18.3. Monotonic algebras

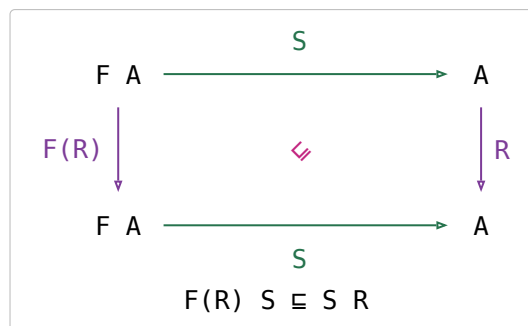
An  $F$ -algebra  $S : F A \rightarrow A$  is **monotonic on**  $R : A \rightarrow A$  if  $F(R) S \sqsubseteq S R$ .

(18.3a)

For a map  $f : F A \rightarrow A$  that is  $f \circ F(R) f \sqsubseteq R$ , equivalently  $F(R) \sqsubseteq f R f^\circ$ .

$(\leq \times \leq) + \sqsubseteq + \leq$  — addition on  $\text{Nat}$  is monotonic on  $\leq$ , which at the point level reads  $c = a + b \wedge a \leq a' \wedge b \leq b' \implies c \leq a' + b'$ .

$F(R) S \sqsubseteq S R$  is (18.2a)'s inequation at  $G := F$ ,  $F := \text{Id}$ ,  $\varphi := S$ , so (18.3b) is (18.2b) with  $\text{Id}(R)$  written  $R$ .

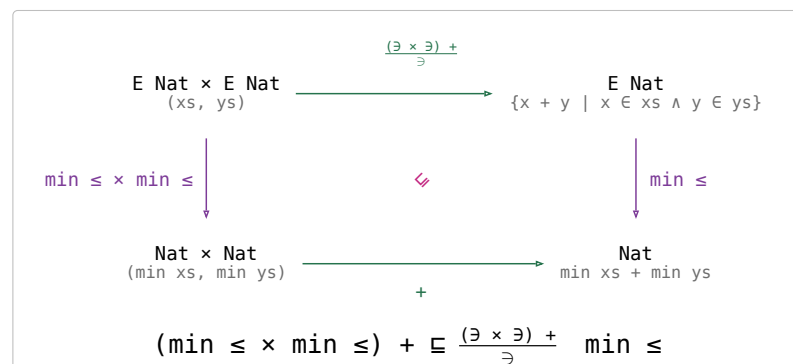
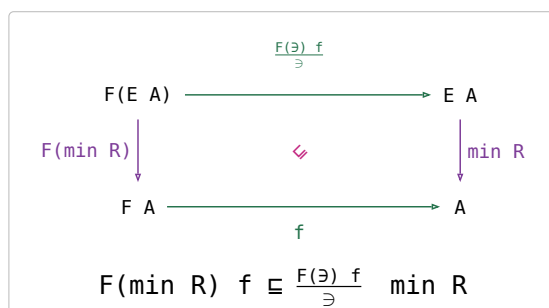


(18.3b)

$f : F A \rightarrow A$  **distributes over**  $R$  if  $F(\min R) f \sqsubseteq \frac{F(\exists) f}{\exists} \min R$ .

(18.3c)

$+$  distributes over  $\leq$ , at the point level  $\min xs + \min ys = \min\{x + y \mid x \in xs \wedge y \in ys\}$  for  $xs, ys$  non-empty and  $\min \triangleq \min \leq$ .



(18.3d)

| the law                                                                                                                                                     | what it says                                                                                                   | (18.3e) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------|
| $f^\circ F(R) f \sqsubseteq R \iff f^\circ F(R^\circ) f \sqsubseteq R^\circ$                                                                                | a <b>map</b> is monotonic on an order and on its opposite together — a relation is not                         |         |
| $f^\circ F(R) f \sqsubseteq R \iff F(\min R) f \sqsubseteq \frac{F(\exists) f}{\exists} \min R$<br>Theorem 7.1, $f$ a map                                   | for a map the two definitions above are one condition                                                          |         |
| $\langle \frac{S}{\exists} \min R \rangle \sqsubseteq \langle \frac{S}{\exists} \rangle \min R$<br>Theorem 7.2, $R$ a preorder and $F(R) S \sqsubseteq S R$ | the <b>greedy theorem</b> : keeping one minimum at every step beats no more than taking the minimum at the end |         |

## §18.4. Planning a company party

(18.4a)

tree A ::= node (A, list(tree A)), base functor  $F(A, B) = A \times \text{list } B$ , rating : Employee  $\rightarrow$  Real.  
 $\text{cost} \triangleq \text{list}(\text{rating}) \text{ sum}$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ ,  $\text{choose} \triangleq \text{outl} \cup \text{outr}$ .  
 $\text{include} \triangleq (1 \times (\text{list}(\text{outr}) \text{ concat})) \text{ cons}$     $\text{exclude} \triangleq (1 \times (\text{list}(\text{choose}) \text{ concat})) \text{ outr}$   
 $S \triangleq \langle \text{include}, \text{exclude} \rangle$ ,  $\text{party} \triangleq \langle S \rangle \text{ choose} : \text{tree Employee} \rightarrow \text{list Employee}$

| the law                                                                                                                                                                                                                                                     | what it says                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| $\frac{\text{party}}{\supset} \max R$                                                                                                                                                                                                                       | the specification: a guest list of greatest total conviviality                                        |
| $(R^\circ \times R^\circ) \text{ choose} \sqsubseteq \text{choose } R^\circ$                                                                                                                                                                                | <b>choose</b> is monotonic on $R^\circ$ — the first claim                                             |
| $(1 \times \text{list}(R^\circ \times R^\circ)) S \sqsubseteq S (R^\circ \times R^\circ)$                                                                                                                                                                   | <b>S</b> is monotonic on $R^\circ \times R^\circ$ — the second claim                                  |
| $\frac{\text{party}}{\supset} \max R \sqsupseteq \langle \frac{S}{\supset} \max(R \times R) \rangle \frac{\text{choose}}{\supset} \max R$                                                                                                                   | the greedy theorem: one best pair per subtree instead of all parties of the whole tree                |
| $\text{party} = \langle S \rangle \frac{\text{choose}}{\supset} \max R$<br>$\text{include} = (1 \times (\text{list}(\text{outr}) \text{ concat})) \text{ cons}$<br>$\text{exclude} = \text{outr list}(\frac{\text{choose}}{\supset} \max R) \text{ concat}$ | the program, <b>exclude</b> renamed: <b>include</b> was already a map, and <b>exclude</b> becomes one |

## §18.5. Shortest paths on a cylinder

(18.5a)

$F(A, X) = A + A \times X$ ,  $L = \text{list}^+$  with initial algebra  $\alpha : F(A, L A) \rightarrow L A$ ,  $N$  the n-tuple relator.  
 $R \triangleq \text{sum} \leq \text{sum}^\circ$ ,  $\text{setify} : N A \rightarrow E A$ ,  $\text{moves} : N A \rightarrow E(N A)$ ,  $\text{trans} : E(N A) \rightarrow N(E A)$ ,  $\text{zip} : F(N A, N B) \rightarrow N F(A, B)$ ,  $\text{cp} \triangleq \frac{F(1, \exists)}{\supset}$ .  
 $\text{generate} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp } P(\alpha)) : F(N A, N(E(L A))) \rightarrow N(E(L A))$   
 $\text{paths} \triangleq \langle \text{generate} \rangle \text{ setify union} : L N \text{ Nat} \rightarrow E(L \text{ Nat})$

| the law                                                                                                                                                                                       | what it says                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| $\text{paths min } R$                                                                                                                                                                         | the specification: a cheapest path across the cylinder                                     |
| $F(1, \text{min } R) \alpha \sqsubseteq \text{cp } P(\alpha) \text{ min } R$<br>(7.13), Theorem 7.1 at the map $\alpha$ , using $\frac{F(1, \exists)}{\supset} \alpha = \text{cp } P(\alpha)$ | extending every path in a set and then taking a minimum is beaten by extending one minimum |
| $\text{generate } N(\text{min } R) \sqsupseteq F(1, N(\text{min } R)) Q$                                                                                                                      | the fusion condition that defines <b>Q</b> — this is what (7.13) is spent on               |
| $Q = F(1, \text{moves trans } N(\text{min } R)) \text{ zip } N(\alpha)$<br>$= [N(\text{wrap}),$<br>$\quad (1 \times \text{moves trans } N(\text{min } R)) \text{ zip}' N(\text{cons})]$       | <b>generate</b> with the minimum taken inside each tuple component                         |
| $\text{paths min } R \sqsupseteq \langle Q \rangle \text{ setify min } R$                                                                                                                     | the program: one fold, $n$ best paths carried per column                                   |

## §18.6. The security van problem

(18.6a)

$R \triangleq \text{length} \leq \text{length}^\circ$ ,  $\text{ceiling} \triangleq \frac{\text{prefix sum}}{\supset} \max \leq$ ,  $\text{floor} \triangleq \frac{\text{prefix sum}}{\supset} \min \leq$ .  
 $\text{secure}$  the coreflexive on  $x$  with  $\text{bmax}(\text{ceiling } x, \text{ceiling } x - \text{floor } x) \leq N$ ;  $\text{ok}$  the coreflexive on  $(a, xs)$  with  $xs$  non-empty and  $[a] \# \text{head } xs$  secure.  
 $\text{new} \triangleq (\text{wrap} \times 1) \text{ cons}$     $\text{glue} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } (\text{cons} \times 1) \text{ cons}$   
 $\text{old} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } ((\text{cons secure}) \times 1) \text{ cons}$   
 $\text{partition} = \langle [\text{nil}, \text{new} \cup \text{glue}] \rangle$ ,  $S \triangleq [\text{nil}, \text{new} \cup \text{old}]$ ,  $\text{partition list}(\text{secure}) = \langle S \rangle$ .  
 $H \triangleq (\text{head prefix}^\circ \text{ head}^\circ) \cup (\text{nil}^\circ \text{ nil})$ ,  $R ; H \triangleq R \cap (R^\circ \Rightarrow H)$ ,  $|R| \triangleq R \cap \neg R^\circ$ .



| the law                                                                                                                                                                                                       | what it says                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| $\text{partition } \text{list}(\text{secure}) \sqsubseteq \min R$                                                                                                                                             | the specification: fewest secure segments                                                                                |
| $\text{secure prefix} \sqsubseteq \text{prefix secure}$                                                                                                                                                       | secure is prefix-closed — the only property of it used until the program                                                 |
| $(1 \times R) \text{ new} \sqsubseteq (\text{new} \cup \text{old}) R$<br>(7.14)                                                                                                                               | starting a new segment is monotonic on R                                                                                 |
| $(1 \times R) \text{ old} \sqsubseteq (\text{new} \cup \text{old}) R$<br>(7.15), FALSE                                                                                                                        | extending the first segment is not — the shorter partition need not stay secure                                          |
| $(1 \times (R ; H)) \text{ new} \sqsubseteq (\text{new} \cup \text{old}) (R ; H)$ (7.16)<br>$(1 \times (R ; H)) \text{ old} \sqsubseteq (\text{new} \cup \text{old}) (R ; H)$ (7.17)                          | both halves do become monotonic once R is refined by H, which breaks a tie in length by the first segment's prefix order |
| $(1 \times T) \text{ new} \sqsubseteq \text{new } H$ (7.18)                                                                                                                                                   | whatever the tail, new's first segment is [a], a prefix of every first segment — this is what (7.16) rests on            |
| $(1 \times T) \text{ old} \sqsubseteq \text{new } H$ (7.19)<br>$(1 \times  R ) \text{ old} \sqsubseteq \text{new } R$ (7.20)<br>$(1 \times (R \cap H)) \text{ old} \sqsubseteq \text{old } (R \cap H)$ (7.21) | the three claims (7.17) rests on, split along<br>$(R ; H) =  R  \cup (R \cap H)$                                         |
| $\text{old} \sqsubseteq \text{new } (R ; H)^\circ$                                                                                                                                                            | old returns a shorter result than new if it returns any result at all, so the fold's choice refines to a map             |
| $\langle [\text{nil}, (\text{ok} \rightarrow \text{glue}, \text{new})] \rangle \sqsubseteq \langle \bigcup \min(R ; H) \rangle$                                                                               | the program: glue onto the first segment while it stays secure, else start a new one                                     |

## §19. Thinning Algorithms

### §19.1. Thinning

For  $Q : A \rightarrow A$ ,  $\text{thin } Q \triangleq (\exists/\exists) \cap (\exists \setminus (Q^\circ \exists)) : E A \rightarrow E A$  (8.1).

$$ys \text{ (thin } Q) \text{ } xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \text{ } Q \text{ } a)$$

(19.1a)

| the law                                                                                                                                                            | what it says                                                                                                              | (19.1b) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------|
| $X \subseteq \frac{S}{\exists} \text{ thin } Q \iff X \exists \subseteq S \text{ and } S^\circ X \subseteq Q^\circ \exists$                                        | the universal property: everything kept is an $S$ -value, and every $S$ -value has a $Q$ -lower bound among the kept ones |         |
| $Q \subseteq R \implies \text{thin } Q \subseteq \text{thin } R$                                                                                                   | the fewer pairs $Q$ relates, the fewer subsets count as thinnings                                                         |         |
| $1 \subseteq \text{thin } Q$ , and $\text{thin } Q$ is a preorder if $Q$ is                                                                                        | keeping everything is always a legal thinning                                                                             |         |
| $\min R = \text{thin } Q \min R$ $Q \subseteq R$ , both preorders                                                                                                  | <b>thin-introduction</b> : thinning first cannot lose an $R$ -minimum                                                     |         |
| $\text{thin } Q \supseteq \min Q \frac{1}{\exists}$ (8.2)                                                                                                          | <b>thin-elimination</b> : keeping one element is a thinning, but its domain is the sets $\min Q$ is defined on            |         |
| $\frac{S}{\exists} \text{ thin } Q \supseteq \frac{S}{\exists} \min R \frac{1}{\exists}$<br>(8.3), $R \cap (S^\circ S) \subseteq Q$                                | the usable variant: $R$ need only refine $Q$ between values $S$ gives one argument                                        |         |
| $\text{union thin } Q \supseteq P(\text{thin } Q) \text{ union}$ (8.4)                                                                                             | thinning each member set is a thinning of the union                                                                       |         |
| $\langle \frac{F(\exists) S}{\exists} \text{ thin } Q \rangle \subseteq \langle \frac{S}{\exists} \rangle \text{ thin } Q$<br>Theorem 8.1, $S$ monotonic on $Q$    | the <b>thinning theorem</b> : thinning at every step beats thinning only at the end                                       |         |
| $\langle \frac{F(\exists) S}{\exists} \text{ thin } Q \rangle \min R \subseteq \langle \frac{S}{\exists} \rangle \min R$<br>Corollary 8.1, $Q \subseteq R$ as well | the same against the optimisation problem itself, by thin-introduction                                                    |         |

### §19.2. Paths in a layered network

$F(A, X) = A + A \times X$ ,  $L = \text{list}^+$  with initial algebra  $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, L A) \rightarrow L A$ .

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$ ,  $\text{consw } (a, (xs, n)) = \langle \text{cons } (a, xs), \text{wt } (a, \text{head } xs) + n \rangle$ .

$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \text{ outr}$ ,  $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

$Q \triangleq R \cap (\text{head head}^\circ)$ ,  $S \triangleq F(1, \exists) \alpha$ ,  $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$ ,  $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$ ,  $\text{step} \triangleq \text{cpr } P(\text{cons}) \min R$ .

(19.2a)

| the law                                                                                                                                                                                                               | what it says                                                                                                             | (19.2b) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------|
| $\langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \min R = \langle \frac{L(\exists)}{\exists} \rangle \min R$                                                                                                     | the specification: a least-cost path across the network                                                                  |         |
| $F(\exists, Q) \alpha \subseteq F(\exists, 1) \alpha Q$                                                                                                                                                               | $S$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                |         |
| $\langle \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin } Q \rangle \min R \subseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \min R$<br>Corollary 8.1                                            | thinning applies: one partial path per starting vertex is enough                                                         |         |
| $S \text{ head} \subseteq [1, \text{outr}]$<br>$S \text{ head}$ simple, so $R \cap (S^\circ S) \subseteq Q$                                                                                                           | the side condition of (8.3): between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |         |
| $\frac{F(\exists, \exists) \alpha}{\exists} \text{ thin } Q$<br>$\supseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} P(\alpha) \min R)$<br>(8.4), (8.3), $P(\frac{1}{\exists}) \text{ union} = 1$ | thin eliminated: split the algebra at $F(\exists, 1) F(1, \exists)$ , thin each part, one minimum per part               |         |
| $\frac{F(1, \exists)}{\exists} P(\alpha) \min R = [\text{wrap}, \text{step}]$                                                                                                                                         | that inner part read off the two branches of $\alpha$                                                                    |         |
| $\langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \min R \supseteq \langle [P(\text{wrap}), \text{cpl } P(\text{step})] \rangle \min R$                                                                           | the program: one fold, a best path per vertex of the current layer                                                       |         |

### §19.3. Implementing thin

(19.3a)

setify : list A  $\rightarrow$  E A, cup : E A  $\times$  E A  $\rightarrow$  E A,  $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$ ,  $\text{listcp}(F) : F(\text{list A}) \rightarrow \text{list}(F A)$ ,  $\text{sort P} \triangleq \text{setify}^\circ \text{ ordered P}$  for P a connected preorder.

thinlist Q is any thinlist Q  $\sqsubseteq$  subseq with thinlist Q setify  $\sqsubseteq$  setify thin Q; one is  $\langle [\text{nil}, \text{bump Q}] \rangle$ ,  $\text{bump Q } (a, []) = [a]$ ,  $\text{bump Q } (a, [b] \# xs) = (b \text{ Q } a \rightarrow [a] \# xs, a \text{ Q } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$ .

**Binary thinning** data:  $S = (f_1 \text{ } p_1) \cup (f_2 \text{ } p_2)$  with  $p_1, p_2$  coreflexive; Q a preorder with  $Q \sqsubseteq R$  and both  $f_1 \text{ } p_1, f_2 \text{ } p_2$  monotonic on Q; P a connected preorder with both  $f_1, f_2$  monotonic on P;  $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$ .

| the law                                                                                                                                                                                          | what it says                                                                                            | (19.3b) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------|
| $\text{thinlist Q } xs = [\text{minlist Q } xs]$<br>(8.5), Q connected, XS non-empty                                                                                                             | what thinning should come to when it can: one element                                                   |         |
| $\text{sort P thinlist Q} \sqsubseteq \text{thin Q sort P}$ (8.6)                                                                                                                                | thinning a sorted list is a thinning of the set — this is what thinlist Q $\sqsubseteq$ subseq buys     |         |
| $\text{sort P minlist Q} \sqsubseteq \text{min Q}$ (8.7)                                                                                                                                         | a minimum of the sorted list is a minimum of the set                                                    |         |
| $\text{sort}(f \text{ P } f^\circ) \text{ list}(f) \sqsubseteq \text{P}(f) \text{ sort P}$ (8.8)                                                                                                 | shunt a function through a sort                                                                         |         |
| $\text{sort P filter}(p) \sqsubseteq \text{E}(p) \text{ sort P}$ (8.9), p coreflexive                                                                                                            | filtering a sorted list sorts the restricted set                                                        |         |
| $(\text{sort P} \times \text{sort P}) \text{ merge P} \sqsubseteq \text{cup sort P}$ (8.10)                                                                                                      | merging two sorted lists sorts their union                                                              |         |
| $F(\text{sort P}) \text{ listcp}(F) \sqsubseteq \text{cp}(F) \text{ sort}(F \text{ P})$<br>(8.11), F linear                                                                                      | listcp(F) is the list implementation of the cartesian product cp(F)                                     |         |
| $F(\text{sort P}) \text{ listcp}(F) \text{ list}(f) \text{ filter}(p) \sqsubseteq \frac{F(\exists) \text{ } f \text{ } p}{\exists} \text{ sort P}$<br>Lemma 8.1, f monotonic on P, p coreflexive | one sorted list built from sorted arguments, instead of a set built and then sorted                     |         |
| $\langle \text{listcp}(F) \langle g_1, g_2 \rangle \text{ merge P thinlist Q} \rangle \text{ minlist R}$<br>$\sqsubseteq \frac{\langle S \rangle}{\exists} \text{ min R}$<br>Theorem 8.2         | the <b>binary thinning theorem</b> : a fold on sorted lists of partial solutions, thinned at every step |         |

### §19.4. The knapsack problem

(19.4a)

vol, wt : Item  $\rightarrow$  Real, value  $\triangleq \text{list}(\text{vol}) \text{ sum}$ , weight  $\triangleq \text{list}(\text{wt}) \text{ sum}$ .

subseq =  $\langle [\text{nil}, \text{cons}] \cup [\text{nil}, \text{outr}] \rangle$ , within w the coreflexive on xs with weight  $xs \leq w$ .

$\geq \triangleq \leq^\circ$ ,  $R \triangleq \text{value} \geq \text{value}^\circ$ ,  $Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ)$ ,  $P \triangleq R$ .

$F A = 1 + \text{Item} \times A$ ,  $\text{listcp}(F) = \text{wrap} + \text{cpr}$ ,  $g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \text{ filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$ ,  $g_2 \triangleq \text{list}([\text{nil}, \text{outr}]) = [\text{list}(\text{nil}), h_2]$ .

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w)$ ,  $h_2 \triangleq \text{list}(\text{outr})$ .

| the law                                                                                                                                                             | what it says                                                                                              | (19.4b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{subseq } (\text{within } w)}{\exists} \text{ min R}$                                                                                                   | the specification: a selection of greatest total value that stays within the capacity w                   |         |
| $\text{subseq } (\text{within } w) = \langle ([\text{nil}, \text{cons}] (\text{within } w)) \cup [\text{nil}, \text{outr}] \rangle$<br>fusion, weights non-negative | the selections that fit are themselves a fold, and the fold's algebra already has binary thinning's shape |         |
| $(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$<br>FALSE                                                             | not monotonic on R: a selection of greater value need not still fit once one more item goes in            |         |
| $(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$<br>$(1 \times Q) \text{ outr} \sqsubseteq \text{outr } Q$            | both halves are monotonic on Q once ties in value are broken by weight                                    |         |
| $\langle \text{listcp}(F) \langle g_1, g_2 \rangle \text{ merge R thinlist Q} \rangle \text{ minlist R}$<br>Theorem 8.2 at $P \triangleq R$ , F linear              | binary thinning, sorting in descending order of value                                                     |         |
| $\text{knapsack } w = \langle [\text{nil}, \text{cpr } \langle h_1, h_2 \rangle \text{ merge R thinlist Q}] \rangle \text{ minlist R}$                              | the program; minlist R becomes head, since packings come out in descending value                          |         |

## §19.5. The paragraph problem

(19.5a)

$\text{Line} = \text{list}^+ \text{ Word}, \text{ Para} = \text{list}^+ \text{ Line}, F A = \text{Word} + \text{Word} \times A, \text{listcp}(F) = \text{wrap} + \text{cpr}.$   
 $\text{new } (a, xs) = [[a]] \# xs, \text{glue } (a, xs) = [[a] \# \text{head } xs] \# \text{tail } xs, \text{partition} \triangleq \lambda ([\text{wrap wrap}, \text{new } u \text{ glue}]) : \text{list}^+ \text{ Word} \rightarrow \text{Para}.$   
 $\text{width} \triangleq \lambda ([\text{length}, (\text{length} \times 1) \text{ plus succ}]), \text{fits } w \text{ the coreflexive on a line } x \text{ with width } x \leq w, \text{ok } w \text{ the coreflexive on } [x] \# xs \text{ with width } x \leq w.$   
 $\text{white } w \ x = w - \text{width } x, \text{collect} \triangleq \text{list}(\text{sqr } \text{sum}), \text{waste } w \triangleq \text{init list}(\text{white } w) \text{ collect}.$   
 $R \triangleq (\text{waste } w) \leq (\text{waste } w)^\circ, Q \triangleq R \cap (\text{head head}^\circ), P \triangleq \top.$   
 $g_1 \triangleq \text{list}([\text{wrap wrap}, \text{new}]), g_2 \triangleq \text{list}([\text{wrap wrap}, \text{glue}]) \text{ filter}(\text{ok } w), \text{start} \triangleq \text{wrap wrap wrap}, h_1 \triangleq \text{list}(\text{new}), h_2 \triangleq \text{list}(\text{glue}) \text{ filter}(\text{ok } w).$

| the law                                                                                                                                                                                                 | what it says                                                                                                             | (19.5b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{partition list}^+(\text{fits } w)}{\ni} \min R$                                                                                                                                            | the specification: least waste among the paragraphs whose every line fits                                                |         |
| $\text{partition list}^+(\text{fits } w) = \lambda ([\text{wrap wrap}, \text{new } u (\text{glue } (\text{ok } w))])$<br>fusion, every word fits on a line by itself                                    | the fitting paragraphs are themselves a fold                                                                             |         |
| $[\text{wrap wrap}, \text{new } u (\text{glue } (\text{ok } w))]$<br>$= [\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] (\text{ok } w))$                                           | rewritten into binary thinning's shape $(f_1 \ p_1) \cup (f_2 \ p_2)$                                                    |         |
| $(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$                                                                                                                               | glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |         |
| $(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$<br>$(1 \times Q) (\text{glue } (\text{ok } w)) \sqsubseteq \text{glue } (\text{ok } w) Q$<br>cons monotonic on collect $\leq \text{collect}^\circ$ | both halves are monotonic on Q once ties in waste are broken by the first line                                           |         |
| merge $\top = \text{cat}; P \triangleq \text{head prefix head}^\circ$ also serves                                                                                                                       | $\top$ needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input                   |         |
| $\lambda (\text{listcp}(F) \ (g_1, g_2) \text{ cat thinlist } Q) \text{ minlist } R$<br>Theorem 8.2 at $P \triangleq \top$                                                                              | binary thinning, one candidate kept per first line                                                                       |         |
| paragraph $w = \lambda ([\text{start}, \text{cpr } (h_1, h_2) \text{ cat thinlist } Q]) \text{ minlist } R$                                                                                             | the program, $g_1$ and $g_2$ split along the coproduct                                                                   |         |

## §19.6. Bitonic tours

(19.6a)

$F A = (\text{City} \times \text{City}) + (\text{City} \times A)$ , the base functor of cons-lists of length at least two;  $\text{listcp}(F) = \text{wrap} + \text{cpr};$   
 $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$ , neither positive nor symmetric.  
 $\text{start } (a, b) = ([a, b], [a, b]), \text{dropl } (a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys), \text{dropr } (a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys), \text{tour} \triangleq \lambda ([\text{start}, \text{dropl } u \text{ dropr}]).$   
 $\text{cost } (xs, ys) = \text{outcost } xs + \text{incost } ys, \text{outcost } [a_0, \dots, a_n] = \text{tc } (a_0, a_1) + \dots + \text{tc } (a_{n-1}, a_n), \text{incost } [a_0, \dots, a_n] = \text{tc } (a_1, a_0) + \dots + \text{tc } (a_n, a_{n-1}).$   
 $\text{next} \triangleq \text{tail head}, \text{next2} \triangleq \text{next} \times \text{next}, R \triangleq \text{cost} \leq \text{cost}^\circ, Q \triangleq R \cap (\text{next2 next2}^\circ), P \triangleq \top, g_1 \triangleq \text{list}([\text{start}, \text{dropl}]), g_2 \triangleq \text{list}([\text{start}, \text{dropr}]).$

| the law                                                                                                                                                    | what it says                                                                                                                  | (19.6b) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{tour}}{\ni} \min R$                                                                                                                           | the specification: a least-cost bitonic tour, outward journey and return kept as a pair of lists                              |         |
| $(1 \times R) \text{ dropl} \sqsubseteq \text{dropl } R \quad \text{FALSE}$<br>$(1 \times R) \text{ dropr} \sqsubseteq \text{dropr } R \quad \text{FALSE}$ | neither drop is monotonic on <b>R</b> : the two edges it adds and removes depend on <b>head</b> and <b>next</b> of both lists |         |
| $(1 \times Q) \text{ dropl} \sqsubseteq \text{dropl } Q$<br>$(1 \times Q) \text{ dropr} \sqsubseteq \text{dropr } Q$                                       | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input            |         |
| $(\text{listcp}(F) \langle g_1, g_2 \rangle \text{ cat thinlist } Q) \text{ minlist } R$<br>Theorem 8.2 at $P \triangleq T$ , merge $T = \text{cat}$       | binary thinning, one candidate per pair of second cities                                                                      |         |
| $\text{mintour} = ([\text{start wrap, cpr } (\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{ cat thinlist } Q]) \text{ minlist } R$           | the program: quadratic, because each step adds just two tours to the list kept                                                |         |

## §20. Dynamic Programming

### §20.1. Theory

$h : F B \rightarrow B$  a map,  $T : F A \rightarrow A$  an  $F$ -algebra,  $R : B \rightarrow B$ .

(20.1a)

$H \triangleq (T)^\circ (h) : A \rightarrow B$ ,  $M \triangleq \frac{H}{\supset} \min R$  the problem to be solved,  $(\mu X : G(X))$  the least fixed point of  $G$ .

| the law                                                                                                                                                                                                                                               | what it says                                                                                        | (20.1b) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------|
| $(\mu X : \frac{T^\circ}{\supset} P(F(X) h) \min R) \sqsubseteq M$<br>Theorem 9.1, $h$ monotonic on $R$                                                                                                                                               | <b>dynamic programming:</b> decompose in every way, solve each part, keep one optimum per part      |         |
| $\frac{T^\circ}{\supset} P(F(M) h) \min R \sqsubseteq M$ (9.1)                                                                                                                                                                                        | all Knaster–Tarski leaves to prove                                                                  |         |
| $\frac{T^\circ}{\supset} P(F(M) h) \min R \sqsubseteq H$ (9.2)<br>$H^\circ \frac{T^\circ}{\supset} P(F(M) h) \min R \sqsubseteq R$ (9.3)                                                                                                              | (9.1) split by the universal property of $\min$                                                     |         |
| $P(X) \min R \sqsubseteq (\exists X) n (E \setminus (X R^\circ))$<br>(9.4) = (7.10), (18.1b)                                                                                                                                                          | the only fact about $\min$ either proof uses                                                        |         |
| $(\mu X : \frac{T^\circ}{\supset} \text{thin } Q P(F(X) h) \min R) \sqsubseteq M$<br>Theorem 9.2, $Q$ a preorder with $Q F(H) h \sqsubseteq F(H) h R$ ; $\text{thin } Q$ as in (19.1b)                                                                | the same with a thinning step: decompositions that can never win are dropped                        |         |
| the fixed point is unique, and entire<br>Theorem 6.3, $T^\circ$ followed by $F$ 's membership relation inductive; $\frac{T^\circ}{\supset}$ finite and non-empty, $R$ connected                                                                       | when the recursion can be refined to a recursive function                                           |         |
| $\frac{[V_1, V_2]^\circ}{\supset} \text{thin}(Q_1 + Q_2) P([U_1, U_2]) \min R$<br>= $(\text{Ran } V_1 \rightarrow W_1, W_2)$<br>$W_i \triangleq \frac{V_i^\circ}{\supset} \text{thin}(Q_i) P(U_i) \min R$<br>Proposition 9.1, $V_2 V_1^\circ = \perp$ | $F A$ is usually a coproduct, and disjoint ranges split the fixed point into one branch per summand |         |
| $F(R) h \sqsubseteq h R$<br>$Q F(H) h \sqsubseteq F(H) h R$                                                                                                                                                                                           | the two conditions to be checked, in the order the three propositions below discharge them          |         |
| $F(R) h \sqsubseteq h R$<br>Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$ , $h \text{ cost} = F(\text{cost}) k$ , $F(\leq) k \sqsubseteq k \leq$                                                                                 | monotonicity when the cost is itself a fold with a step $k$ monotonic on $\leq$                     |         |
| $F(R n (H^\circ H)) h \sqsubseteq h R$<br>Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$ , $h \text{ cost} = F(\langle \text{cost}, H^\circ \rangle) k$ , $F(\leq \times \mathbb{1}) k \sqsubseteq k \leq$ , $H^\circ$ simple     | monotonicity <b>in context</b> : $k$ may also read the input the part was built from                |         |
| $Q \triangleq F(U, V)$<br>Proposition 9.4, $U, V$ preorders, $F(U, R) h \sqsubseteq h R$ , $V H \sqsubseteq H R$                                                                                                                                      | both conditions of Theorem 9.2 at once, split along the two arguments of a bifunctor                |         |

### §20.2. The string edit problem

$\text{Op} ::= \text{cpy Char} \mid \text{del Char} \mid \text{ins Char}$ ,  $F(A, B) = 1 + (A \times B)$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ .

(20.2a)

$\text{edit} \triangleq ([\text{base}, \text{step}]) : \text{list Op} \rightarrow \text{list Char} \times \text{list Char}$ ,  $\text{base}$  returning  $([], [])$ ,  $\text{step}(\text{cpy } a, (xs, ys)) = ([a] \# xs, [a] \# ys)$ ,  $\text{step}(\text{del } a, (xs, ys)) = ([a] \# xs, ys)$ ,  $\text{step}(\text{ins } a, (xs, ys)) = (xs, [a] \# ys)$ .

$\text{length} \triangleq ([\text{zero}, \text{outr succ}])$ ,  $R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U \triangleq \tau$ ,  $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$ ,  $Q \triangleq \mathbb{1} + (U \times V)$ , empty the coreflexive on  $(xs, ys)$  with both lists empty.

$\text{unstep}$  implements  $\frac{\text{step}^\circ}{\supset} \text{thin}(U \times V)$ :  $\text{unstep}([a] \# xs, []) = [(\text{del } a, (xs, []))]$ ,  $\text{unstep}([], [b] \# ys) = [(\text{ins } b, ([], ys))]$ ,  $\text{unstep}([a] \# xs, [b] \# ys) = (a = b \rightarrow [(\text{cpy } a, (xs, ys))], [(\text{del } a, (xs, [b] \# ys)), (\text{ins } b, ([a] \# xs, ys))])$ .

| the law                                                                                                                                                                                                                                                                                                                                      | what it says                                                                                                                                          | (20.2b) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{edit}^\circ}{\supset} \min R$                                                                                                                                                                                                                                                                                                   | the specification: a shortest edit sequence from which both strings can be reconstituted                                                              |         |
| $F(R) \alpha \sqsubseteq \alpha R$<br>Proposition 9.2 at <code>length</code> , <code>succ</code> monotonic on $\leq$                                                                                                                                                                                                                         | <code>cons</code> is monotonic on $R$ , so Theorem 9.1 already applies                                                                                |         |
| $Q F(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R$                                                                                                                                                                                                                                                              | the thinning condition sought, over $F(0p, \text{list Char} \times \text{list Char})$                                                                 |         |
| $F(T, R) \alpha \sqsubseteq \alpha R$ left as an exercise                                                                                                                                                                                                                                                                                    | $U \triangleq T$ costs nothing: any two operations may be compared                                                                                    |         |
| $\text{edit } (1 \times \text{suffix}) \sqsubseteq R^\circ \text{ edit}$<br>$\text{edit } (\text{suffix} \times 1) \sqsubseteq R^\circ \text{ edit}$                                                                                                                                                                                         | the other half of Proposition 9.4 for $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$                                                   |         |
| $\text{edit } (1 \times \text{tail}) \sqsubseteq R^\circ \text{ edit}$<br>$\text{edit } (\text{tail} \times 1) \sqsubseteq R^\circ \text{ edit}$<br>$\text{suffix} = \text{tail}^*$ , and $B A \sqsubseteq C B \implies B A^* \sqsubseteq C^* B$                                                                                             | one step is enough: drop the operation that produced the head, or weaken its <code>cpy</code> to a <code>del</code> — never lengthening the sequence  |         |
| $(\mu X : \frac{[\text{base}, \text{step}]^\circ}{\supset} \text{thin } Q P([\text{nil}, (1 \times X) \text{ cons}]) \min R) \sqsubseteq \frac{\text{edit}^\circ}{\supset} \min R$<br>Theorem 9.2                                                                                                                                            | a copy, when available, beats a delete or an insert                                                                                                   |         |
| $X = (\text{empty} \rightarrow \text{nil}, \frac{\text{step}^\circ}{\supset} \text{thin}(U \times V) P((1 \times X) \text{ cons}) \min R)$ Proposition 9.1                                                                                                                                                                                   | <code>base</code> and <code>step</code> have disjoint ranges                                                                                          |         |
| $\text{mle} = (\text{empty} \rightarrow \text{nil}, \text{unstep list}((1 \times \text{mle}) \text{ cons}) \text{minlist } R)$                                                                                                                                                                                                               | the program: at most two decompositions survive, but the same subproblem is solved many times, so the running time is exponential                     |         |
| $\text{column } xs \text{ } ys = [\text{mle } (u, ys) \mid u \leftarrow \text{tails } xs]$<br>$\text{column } xs = ([\text{fstcol } xs, \text{nextcol } xs]), \text{fstcol} = \text{list}(\text{del}) \text{ tails}$                                                                                                                         | the tabulation: $\text{mle } (xs, ys)$ needs $\text{mle } (u, v)$ for every tail $u$ of $xs$ and $v$ of $ys$ , so the columns are built right to left |         |
| $\text{column } xs \text{ } ([b] \# ys) = \text{nextcol } xs \text{ } (b, \text{column } xs \text{ } ys)$<br>$\text{nextcol } xs \text{ } (b, us) = ([[\text{base } (b, \text{last } us), \text{step } b]] \# xus$<br>$xus = \text{zip } (xs, \text{zip } (\text{init } us, \text{tail } us))$                                               | each column is a fold built bottom to top, over $xs$ zipped with the adjacent pairs of the column to its right                                        |         |
| $\text{base } (b, u) = [[\text{ins } b] \# u]$<br>$\text{step } b \text{ } ((a, (u, v)), ws) =$<br>$(a = b \rightarrow [[\text{cpy } a] \# v] \# ws,$<br>$[\text{bmin } R \text{ } ([\text{del } a] \# w, [\text{ins } b] \# u)] \# ws)$<br>$w = \text{head } ws$ ; these <code>base</code> , <code>step</code> are not <code>edit</code> 's | an entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths     |         |

### §20.3. Optimal bracketing

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <p><code>tree A ::= tip A   bin (tree A, tree A), F X = A + X<sup>2</sup>, so F(R) = 1 + R<sup>2</sup>; h <math>\triangleq</math> [tip, bin], flatten <math>\triangleq</math> ([wrap, cat]) : tree A <math>\rightarrow</math> list<sup>+</sup> A, H = flatten<sup>o</sup>.</code></p> <p>(cost, size) <math>\triangleq</math> ([opt, opb]), opt <math>\triangleq</math> (zero, st), opb ((cx, sx), (cy, sy)) = (cb (sx, sy) + cx + cy, sb (sx, sy)).</p> <p>sb associative, so size = flatten sz for a map sz; R <math>\triangleq</math> cost <math>\leq</math> cost<sup>o</sup>, g <math>\triangleq</math> [zero, (1 <math>\times</math> sz)<sup>2</sup> opb outl], single the coreflexive on singleton lists.</p> <p>splits <math>\triangleq</math> (inits<sup>+</sup>, tails<sup>+</sup>) zip, an implementation of <math>\frac{\text{cat}^\circ}{\supset}</math>; array <math>\triangleq</math> inits list(row), row <math>\triangleq</math> tails list(mct), col <math>\triangleq</math> inits list(mct).</p> <p>mix <math>\triangleq</math> zip list(bin) minlist R, next <math>\triangleq</math> (outl, mix) snoc, process <math>\triangleq</math> ((tip wrap) <math>\times</math> 1) loop(next).</p> | (20.3a) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|

| the law                                                                                                                                                                                                                                                                                                                             | what it says                                                                                                              | (20.3b) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{flatten}^\circ}{\ni} \min R$                                                                                                                                                                                                                                                                                           | the specification: a least-cost bracketing of $a_1 \oplus \dots \oplus a_n$ , the tree whose flattening is the given list |         |
| $[\text{tip}, \text{bin}] \text{ cost} = (1 + (\text{cost}, \text{flatten})^2) g \quad (9.5)$                                                                                                                                                                                                                                       | the cost of a node reads only the cost and the flattening of its two subtrees                                             |         |
| $(1 + (\leq \times 1)^2) g \sqsubseteq g \leq \quad (9.6)$                                                                                                                                                                                                                                                                          | $g$ is monotonic on $\leq$ in its two cost arguments                                                                      |         |
| $F(R \cap (\text{flatten} \text{ flatten}^\circ)) h \sqsubseteq h R$<br>Proposition 9.3, $H^\circ = \text{flatten}$ a map                                                                                                                                                                                                           | monotonicity in context: only trees with the same flattening are compared                                                 |         |
| $(\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{\ni} P([\text{tip}, (X \times X) \text{ bin}]) \min R) \sqsubseteq \frac{\text{flatten}^\circ}{\ni} \min R$<br>Theorem 9.1                                                                                                                                                         | split the list in every way, bracket both halves, join                                                                    |         |
| $X = (\text{single} \rightarrow \text{wrap}^\circ \text{ tip}, \frac{\text{cat}^\circ}{\ni} P((X \times X) \text{ bin}) \min R)$<br>Proposition 9.1                                                                                                                                                                                 | $\text{wrap}$ and $\text{cat}$ have disjoint ranges                                                                       |         |
| $\text{mct} = (\text{single} \rightarrow \text{head tip}, \text{splits list}((\text{mct} \times \text{mct}) \text{ bin}) \text{ minlist } R)$                                                                                                                                                                                       | the program; exponential, since the segments of one list overlap                                                          |         |
| $\text{mct} = (\text{single} \rightarrow \text{head tip}, (\text{init col}, \text{tail row}) \text{ mix}) \quad (9.7)$                                                                                                                                                                                                              | the tabulation: $\text{mct } xs$ is needed for every non-empty segment $xs$ , so the values are held as an array of rows  |         |
| $\text{col} = (\text{single} \rightarrow \text{head tip wrap}, (\text{init col}, \text{tail row}) \text{ next}) \quad (9.8)$<br>$\text{cons col} = (1 \times \text{array}) \text{ process} \quad (9.9)$<br>$\text{row} = (\text{single} \rightarrow \text{head tip wrap}, (\text{mct}, \text{tail row}) \text{ cons}) \quad (9.10)$ | a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop                       |         |
| $\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \triangleq \text{tip wrap wrap}$<br>$\text{addcol} \triangleq (\text{outl tip wrap}, \text{step}) \text{ cons}$<br>$\text{step} \triangleq (\text{process tail}, \text{outr}) \text{ zip list}(\text{cons})$                                                        | the program: one fold building the array column by column, cubic in the length of the input                               |         |

## §20.4. Data compression

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <pre>list A ::= nil   snoc (list A, A), list<sup>+</sup> A ::= wrap A   snoc (list<sup>+</sup> A, A), String = list Char, Code ::= sym Char   ptr (String, String<sup>+</sup>), F(Code, String) = 1 + (String × Code), α ≜ [nil, snoc].  decode ≜ ([nil, extend]) : list Code → String, extend (xs, sym a) = xs # [a], extend (xs, ptr (ys, zs)) = xs # zs when ys # zs is a proper prefix of xs # zs; H = decode<sup>°</sup>.  size ≜ ([zero, distr [1 × c, 1 × p] plus]) with c, p the constant costs of a symbol and a pointer; R ≜ size ≤ size<sup>°</sup>, Q ≜ F(τ + τ, prefix<sup>°</sup>) = 1 + (prefix<sup>°</sup> × (τ + τ)), the two τ on symbols and on pointers.  lrt ws = min(prefix<sup>°</sup> × (τ + τ)) {(xs, (ys, zs))   xs # zs = ws, ys # zs a proper prefix of ws}, the longest repeated tail; reduce (ws # [a]) = (zs ≠ [] → [(ws, sym a), (xs, ptr (ys, zs))], [(ws, sym a)]) with (xs, (ys, zs)) = lrt (ws # [a]).</pre> | (20.4a) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|



| the law                                                                                                                                                                                     | what it says                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{\text{decode}^\circ}{\ni} \min R$                                                                                                                                                    | the specification: a smallest code sequence decoding to the given string                                                             |
| $F(R) \alpha \sqsubseteq \alpha R \quad (\tau + \tau) [c, p] = [c, p]$                                                                                                                      | monotonicity is routine, the two costs being constants                                                                               |
| $F(\tau + \tau, R) \alpha \sqsubseteq \alpha R$<br>$\text{decode prefix} \sqsubseteq R^\circ \text{ decode}$<br>Proposition 9.4 at $Q \triangleq F(\tau + \tau, \text{prefix}^\circ)$       | the thinning condition split in two: pointers are ordered only by how much input they consume, symbols and pointers not at all       |
| $\text{decode init} \sqsubseteq R^\circ \text{ decode}$                                                                                                                                     | enough for the second: drop the last character of the output by shortening or removing the last code element, never raising the cost |
| $(\mu X : \frac{[\text{nil}, \text{extend}]^\circ}{\ni} \text{thin } Q P([\text{nil}, (X \times 1) \text{snoc}]) \min R) \sqsubseteq \frac{\text{decode}^\circ}{\ni} \min R$<br>Theorem 9.2 | between a symbol and a pointer nothing can be decided in advance; between two pointers the longer match wins                         |
| $X = (\text{null} \rightarrow \text{nil}, \frac{\text{extend}^\circ}{\ni} \text{thin}(\text{prefix}^\circ \times (\tau + \tau)) P((X \times 1) \text{snoc}) \min R)$<br>Proposition 9.1     | $\text{nil}$ and $\text{extend}$ have disjoint ranges                                                                                |
| $\frac{\text{extend}^\circ}{\ni} (ws \# [a]) = \{(ws, \text{sym } a)\} \cup \{(xs, \text{ptr } (ys, zs)) \mid xs \# zs = ws \# [a], ys \# zs \text{ a proper prefix of } ws\}$              | the decompositions of one string: take the last character as a symbol, or end with a pointer                                         |
| $\text{reduce implements } \frac{\text{extend}^\circ}{\ni} \text{thin}(\text{prefix}^\circ \times (\tau + \tau))$                                                                           | thinning leaves at most two: the symbol, and the pointer of $\text{lrt}$                                                             |
| $\text{encode} = (\text{null} \rightarrow \text{nil}, \text{reduce list}((\text{encode} \times 1) \text{snoc}) \text{minlist } R)$                                                          | the program, again exponential; the book gives no tabulation for it                                                                  |

## §21. Greedy Algorithms

### §21.1. Theory

$h, T, R, H, M$  as in (20.1a); additionally  $Q$  a **connected** preorder on the sets  $\frac{T^\circ}{\supseteq}$  returns, so that  $\frac{T^\circ}{\supseteq} \min Q$  is entire. (21.1a)

| the law                                                                                                                                                                                                          | what it says                                                                                                                                                                                                                                          | (21.1b) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $(\mu X : \frac{T^\circ}{\supseteq} \min Q F(X) h) \sqsubseteq M$<br>Theorem 10.1, the hypotheses of Theorem 9.2                                                                                                 | <b>greedy</b> : one decomposition is kept at every step, so $P$ and $\frac{T^\circ}{\supseteq}$ disappear from the recursion                                                                                                                          |         |
| $\frac{[V_1, V_2]^\circ}{\supseteq} \min(Q_1 + Q_2) [U_1, U_2]$<br>$= (Ran V_1 \rightarrow W_1, W_2)$<br>$W_i \triangleq \frac{V_i^\circ}{\supseteq} \min(Q_i) U_i$<br>Proposition 10.1, $V_2 V_1^\circ = \perp$ | Proposition 9.1 with $\min$ for $\text{thin}$                                                                                                                                                                                                         |         |
| $Q \triangleq F(U, V)$ Proposition 9.4                                                                                                                                                                           | still the way to get both conditions, but $\frac{T^\circ}{\supseteq} \min Q$ must now be entire as well                                                                                                                                               |         |
| Theorem 7.2, (18.3e)                                                                                                                                                                                             | that greedy theorem chooses among the <b>results</b> of one relational step of a catamorphism; Theorem 10.1 chooses among the <b>decompositions</b> of the input, for an arbitrary $T$ rather than an initial algebra, at the cost of $h$ being a map |         |

### §21.2. The detab-entab problem

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$  over snoc-lists,  $\alpha \triangleq [\text{nil}, \text{snoc}]$ ,  $H = \text{detab}^\circ$ ;  $\text{expand}(xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a]), \text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$ . (21.2a)

$\text{col} \triangleq ([\text{zero}, \text{count}])$ ,  $\text{count}(c, a) = (a = \text{NL} \rightarrow 0, c + 1)$ ; TB the tab, BL the blank, NL the newline, tab stops every  $n$  columns.

$R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U$  the preorder with  $a U b \iff a = \text{TB} \vee a = b$ ,  $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$ ,  $Q \triangleq 1 + (V \times U)$ .

$\text{unfill } xs$  the shortest prefix of  $xs$  with  $\text{fill } (\text{unfill } xs) = \text{fill } xs$ ;  $\text{tbc}$  the trailing blank count,  $\text{triple} \triangleq (\text{unfill } \text{entab}, \{\text{tbc}, \text{col}\})$ .

| the law                                                                                                                                                                                                                                                                                                                  | what it says                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{\text{detab}^\circ}{\supset} \min R$                                                                                                                                                                                                                                                                              | the specification: a shortest input <b>detab</b> expands to the given output — <b>detab entab</b> = 1 and nothing shorter does                            |
| $F(T, R) \alpha \sqsubseteq \alpha R$ left as an exercise                                                                                                                                                                                                                                                                | $U$ may be any preorder on characters; $a U b \iff a = TB \vee a = b$ puts <b>TB</b> below every character, so <b>min</b> prefers a <b>tab</b> to a blank |
| $\text{detab prefix} \sqsubseteq R^\circ \text{detab}$ FALSE                                                                                                                                                                                                                                                             | at $n = 8$ , $\text{detab } [a,b,c,d,e,TB] = [a,b,c,d,e,BL,BL,BL]$ , whose prefix $[a,b,c,d,e,BL,BL]$ is longer than any input giving it                  |
| $\text{detab } V^\circ \sqsubseteq R^\circ \text{detab} \quad V \triangleq \text{prefix}^\circ n \text{ (fill fill}^\circ)$                                                                                                                                                                                              | true once only the prefixes that do not cross a tab stop are allowed                                                                                      |
| $\text{nil } V^\circ = \text{nil} \quad \text{fill } V^\circ = \text{fill}$<br>$\text{snoc } V^\circ \sqsubseteq \text{snoc } U \text{ (outl } V^\circ)$ left as exercises                                                                                                                                               | the three properties of $V$ the derivation rests on; the second is what <b>fill fill</b> <sup>°</sup> was added for                                       |
| $\text{expand } V^\circ \sqsubseteq \text{expand } U \text{ (outl } V^\circ)$                                                                                                                                                                                                                                            | the claim they add up to: shortening the output either leaves the last step alone or discards it                                                          |
| $\text{detab } V^\circ \sqsubseteq \text{detab } U \text{ (init detab } V^\circ)$<br><i>init</i> inductive, so the greatest solution is the unique one                                                                                                                                                                   | hence $\text{detab } V^\circ \sqsubseteq \text{prefix detab}$ , and $\text{prefix} \sqsubseteq R^\circ$ finishes Proposition 9.4                          |
| $(\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{\supset} \min Q (1 + (X \times 1)) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{\supset} \min R$<br>Theorem 10.1                                                                                                                                  | greedy: one character of input is decided at each step                                                                                                    |
| $X = (\text{null} \rightarrow \text{nil}, \frac{\text{expand}^\circ}{\supset} \min(V \times U) (X \times 1) \text{snoc})$<br>Proposition 10.1                                                                                                                                                                            | <b>nil</b> and <b>expand</b> have disjoint ranges                                                                                                         |
| $\frac{\text{expand}^\circ}{\supset} (xs \# [a]) = \{(ys, TB) \mid \text{fill } ys = xs \# [a]\} \cup \{(xs, a)\}$<br>the first set is non-empty iff $a = BL$ and $\text{col } (xs \# [a]) \bmod n = 0$                                                                                                                  | the last output character came from a tab only if it is a blank landing exactly on a tab stop                                                             |
| $\frac{\text{expand}^\circ}{\supset} \min(V \times U) (xs \# [a]) = (a = BL \wedge \text{col } (xs \# [a]) \bmod n = 0 \rightarrow (\text{unfill } xs, TB), (xs, a))$                                                                                                                                                    | the greedy step: emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop                                               |
| $\text{entab } xs = \text{entab } (\text{unfill } xs) \# \text{blanks } (\text{tbc } xs) \text{ (10.1)}$                                                                                                                                                                                                                 | what makes <b>triple</b> a <b>snoc</b> -list catamorphism: the output splits at the last tab stop                                                         |
| $\text{triple} = ([\text{base}, \text{op}])$ and $\text{entab} = \text{triple assocl outl } (1 \times \text{blanks}) \text{ cat}$                                                                                                                                                                                        | the program: one pass carrying the column and the count of pending blanks                                                                                 |
| <b>base</b> returns $([], (0, 0))$<br><b>op</b> $((xs, (t, c)), a) =$<br>$(a = BL \wedge (c + 1) \bmod n \neq 0 \rightarrow (xs, (t + 1, c + 1)),$<br>$a = BL \rightarrow (xs \# [TB], (0, c + 1)),$<br>$a = NL \rightarrow (xs \# \text{blanks } t \# [NL], (0, 0)),$<br>$(xs \# \text{blanks } t \# [a], (0, c + 1)))$ | hold a blank back, cash the held blanks in for a tab at a tab stop, flush them at a newline, flush them before anything else                              |

### §21.3. The minimum tardiness problem

$F X = 1 + (X \times \text{Job})$ ,  $\alpha \triangleq [\text{nil}, \text{snoc}]$  on schedules,  $\beta \triangleq [\text{nil}, \text{snag}]$  on bags, **snag** putting a job into a bag;  $\text{bagify} \triangleq ([\beta]) : \text{list Job} \rightarrow \text{Bag Job}$ ,  $H = \text{bagify}^\circ$ .

**ct**, **dt**, **wt** : **Job** → **Real** the completion, due and weighting quantities of a job;  $\text{penalty } (xs, j) = (\text{sum } (\text{list}(\text{ct } xs) + \text{ct } j - \text{dt } j) \times \text{wt } j)$ .

$\text{cost} \triangleq \frac{\text{prefix}}{\supset} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{max} \leq$ ,  $\text{cost } [] = 0$ ,  $\text{cost } (xs \# [j]) = \text{bmax } (\text{cost } xs, \text{penalty } (xs, j))$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

$\text{perm} \triangleq \text{bagify bagify}^\circ = ([\text{nil}, \text{add}])$ ,  $\text{add } (xs, j) = ys \# [j] \# zs$  for some  $xs = ys \# zs$ .

$k \triangleq [\text{zero}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{penalty})) \text{bmax}]$ ,  $f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{penalty}]$ ,  $Q \triangleq f \leq f^\circ$ .

| the law                                                                                                                                                                                                                                | what it says                                                                                                                                                                                       | (21.3b) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{bagify}^\circ}{\ni} \min R$                                                                                                                                                                                               | the specification: an ordering of the given bag of jobs with least maximum penalty                                                                                                                 |         |
| $F(R \cap (\text{bagify } \text{bagify}^\circ)) \alpha \sqsubseteq \alpha R \quad (10.2)$                                                                                                                                              | monotonicity in context — only schedules of one bag are compared                                                                                                                                   |         |
| $(Q \cap (\beta \beta^\circ)) F(\text{bagify}^\circ) \alpha \sqsubseteq F(\text{bagify}^\circ) \alpha R \quad (10.3)$                                                                                                                  | the greedy condition, also in context                                                                                                                                                              |         |
| $\text{cost } (xs \# [j]) = \text{bmax } (\text{cost } xs, \text{penalty } (\text{perm } xs, j))$<br>$\alpha \text{ cost} = F(\langle \text{cost}, \text{bagify} \rangle) k, F(\leq \times 1) k \sqsubseteq k \leq$<br>Proposition 9.3 | (10.2): <b>penalty</b> sums the completion times of $xs$ , and a sum does not see the order                                                                                                        |         |
| $\alpha \text{ cost} = \langle g, h \rangle \text{bmax} \quad (10.4)$<br>$g \triangleq [\text{zero}, \text{penalty}] \quad (10.5)$<br>$h \triangleq [\text{zero}, \text{outl cost}] \quad (10.6)$                                      | the cost of a schedule split into the penalty of its last job and the cost of the rest                                                                                                             |         |
| $\text{add} \sqsubseteq \text{outl } R \quad (10.7)$<br>$\beta \text{ bagify}^\circ = F(\text{bagify}^\circ) [\text{nil}, \text{add}] \quad (10.8)$                                                                                    | adding a job never lowers the cost; and $\text{bagify}^\circ$ is itself a catamorphism on bags. Together: $\beta \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) h \leq \text{cost}^\circ$ |         |
| $F(\text{bagify}) Q^\circ F(\text{bagify}^\circ) = g \geq g^\circ$ , met by<br>$Q \triangleq f \leq f^\circ, f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{penalty}]$                                                | the specification $Q$ has to satisfy, and the choice that meets it: a $Q$ -minimum is a job of least penalty                                                                                       |         |
| no greedy <b>catamorphism</b> exists                                                                                                                                                                                                   | a greedy snoc-list catamorphism would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole                                      |         |
| $X = (\text{null} \rightarrow \text{nil}, \frac{\text{snag}^\circ}{\ni} \min Q (X \times 1) \text{snoc}) \quad \text{Theorem 10.1, Proposition 10.1}$                                                                                  | $\text{nil}$ and $\text{snag}$ have disjoint ranges                                                                                                                                                |         |
| $\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick } (\text{schedule} \times 1) \text{snoc})$<br>$\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{\ni} \min Q$ , a partial function                                   | the program: repeatedly remove a job of least penalty and put it last; quadratic in the number of jobs                                                                                             |         |

## §21.4. The TeX problem

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |
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| <p>intern <math>\triangleq</math> val round : Decimal <math>\rightarrow</math> [0, <math>2^{16}</math>], val <math>\triangleq</math> ([zero, shift]), shift (d, r) = (d + r)/10, round r rounds <math>2^{16} r</math> to the nearest integer: round r = n <math>\iff 2n - 1 &lt; 2^{17} r &lt; 2n + 1</math>.</p> <p>interval n = ((2n - 1)/<math>2^{17}</math>, (2n + 1)/<math>2^{17}</math>), r inrange (a, b) <math>\iff a &lt; r &lt; b</math>, round<math>^\circ</math> = interval inrange, R <math>\triangleq</math> length <math>\leq</math> length<math>^\circ</math>.</p> <p>Interval the pairs (a, b) with <math>0 &lt; b &lt; 1</math> and <math>a &lt; b</math> (10.9) [arb, step] : 1 + (Digit <math>\times</math> Interval) <math>\rightarrow</math> Interval, step (d, (a, b)) = ((d + a)/10, (d + b)/10).</p> <p>F X = 1 + (Digit <math>\times</math> X), <math>\alpha \triangleq</math> [nil, cons], h <math>\triangleq</math> ([arb, step]) = H<math>^\circ</math>, ! : Digit <math>\times</math> Interval <math>\rightarrow</math> 1, Q <math>\triangleq</math> (11<math>^\circ</math> !<math>^\circ</math> 1r) <math>\cup</math> 1, w <math>\triangleq</math> <math>2^{17}</math>.</p> | (21.4a) |
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| the law                                                                                                                                                                                             | what it says                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{\text{intern}^\circ}{\supset} \min R = \text{interval} \frac{\text{inrange val}^\circ}{\supset} \min R$                                                                                      | the specification: a shortest decimal whose internal representation is the given multiple of $2^{-16}$ . $\text{round}^\circ$ is not a map, but $\text{interval}$ is, so it comes out of the $\frac{\supset}{\supset}$ |
| $\text{val inrange}^\circ = ([\text{arb}, \text{step}])$<br>fusion; $\text{zero inrange}^\circ = \text{arb}, \text{shift inrange}^\circ = (1 \times \text{inrange}^\circ) \text{ step}$             | the converse of $\text{val}$ , cut down to intervals, is a catamorphism on cons-lists                                                                                                                                  |
| $F(R) \alpha \sqsubseteq \alpha R$                                                                                                                                                                  | $\text{cons}$ is monotonic on $R$ — routine                                                                                                                                                                            |
| $\alpha^\circ F(h) Q^\circ \sqsubseteq R^\circ \alpha^\circ F(h)$                                                                                                                                   | the greedy condition of Theorem 10.1, converted                                                                                                                                                                        |
| $\frac{\text{arb}^\circ}{\supset} (a, b) = (a < 0 \rightarrow \{*\}, \{\})$<br>$\frac{\text{step}^\circ}{\supset} (a, b) = \{(d, (10a - d, 10b - d)) \mid 0 < 10b - d < 1\}$                        | the decompositions of an interval: stop, or take one more digit                                                                                                                                                        |
| $0 < 10b - d_1 < 1 \text{ and } 0 < 10b - d_2 < 1 \text{ imply } d_1 = d_2$                                                                                                                         | $\text{step}^\circ$ is in fact a map, $d$ the digit with $0 < 10b - d < 1$ , so $\frac{[\text{arb}, \text{step}]^\circ}{\supset}$ returns at most two elements and $Q$ need only choose between them                   |
| $Q \triangleq (\text{!}^\circ \text{!}^\circ \text{!}^\circ) \cup \text{!}$ , and $\text{! nil} \sqsubseteq \text{cons } R^\circ$<br>from $\text{! nil length} \sqsubseteq \text{cons length} \geq$ | stop whenever stopping is legal: the empty decimal is shorter than any other                                                                                                                                           |
| $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{\supset} \min Q F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{\supset} \min R$ Theorem 10.1                                                  | greedy: emit the one digit the interval allows, until the interval contains zero                                                                                                                                       |
| $\text{extern} = \text{interval } f$<br>$f(a, b) = (a < 0 \rightarrow [], [d] \# f(10a - d, 10b - d))$                                                                                              | the program, with $d$ the digit above                                                                                                                                                                                  |
| $\text{extern } n = f(2n - 1, 2n + 1)$<br>$f(p, q) = (p \leq 0 \rightarrow [], [d] \# f(10p - w d, 10q - w d))$<br>$d = (10q) \text{ div } w, w = 2^{17}$                                           | the same in integer arithmetic only, as chapter 3 required of $\text{intern}$ : every interval reached is $(p/w, q/w)$                                                                                                 |